Block practice · Graphs

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Writing down the asymptotes of a rational curve

Where a rational curve crosses the axes

Filling in a table of values

Drawing the graph from your table

Sketching a rational curve with its asymptotes

Sketching curves and where they cross the axes

Solving an equation with your graph and a straight line

Solving log and exponential equations with your graph

OLevels.com SCHOOL

Practice workbook · Writing down the asymptotes of a rational curve

Writing down the vertical and horizontal asymptotes of a curve $y = (ax + b)/(cx + d)$.

19 question parts: 1 worked-example retry and 18 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-GRA-04

1. Question 3 · part d · 1 mark

2021 NOV · Paper 02

Worked-example retry · Attempt part d only.

3

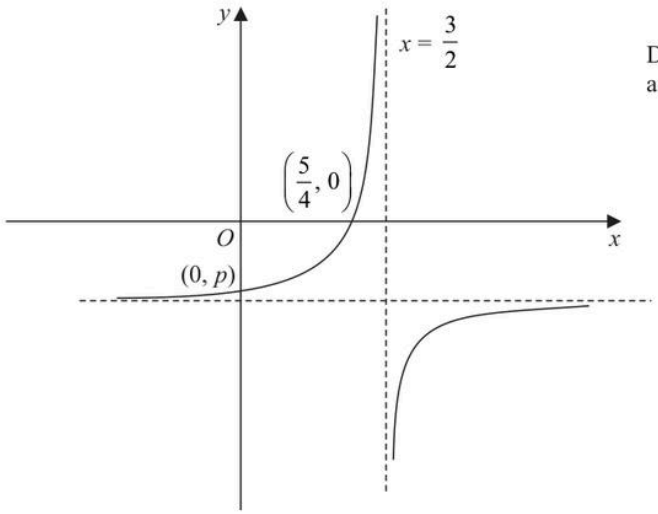


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sketch of the curve with equation

$$y = \frac{a - bx}{cx - d} \qquad x \neq \frac{d}{c}$$

where a, c and d are prime numbers and b is an integer.

The asymptote to the curve that is parallel to the y-axis has equation $x = \frac{3}{2}$

(a) Write down the value of c and the value of d

(2)

The curve crosses the x-axis at the point $\left(\frac{5}{4}, 0\right)$

(b) Find the value of a and the value of b

(2)

The curve crosses the y-axis at the point $(0, p)$ where p is a rational number.

(c) Find the value of p

(2)

(d) Find an equation of the asymptote to the curve that is parallel to the x-axis.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

8

P 6 6 0 2 7 R R A 0 8 3 6

1. Question 3 · part d · 1 mark

2021 NOV · Paper 02

Worked-example retry · Attempt part d only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

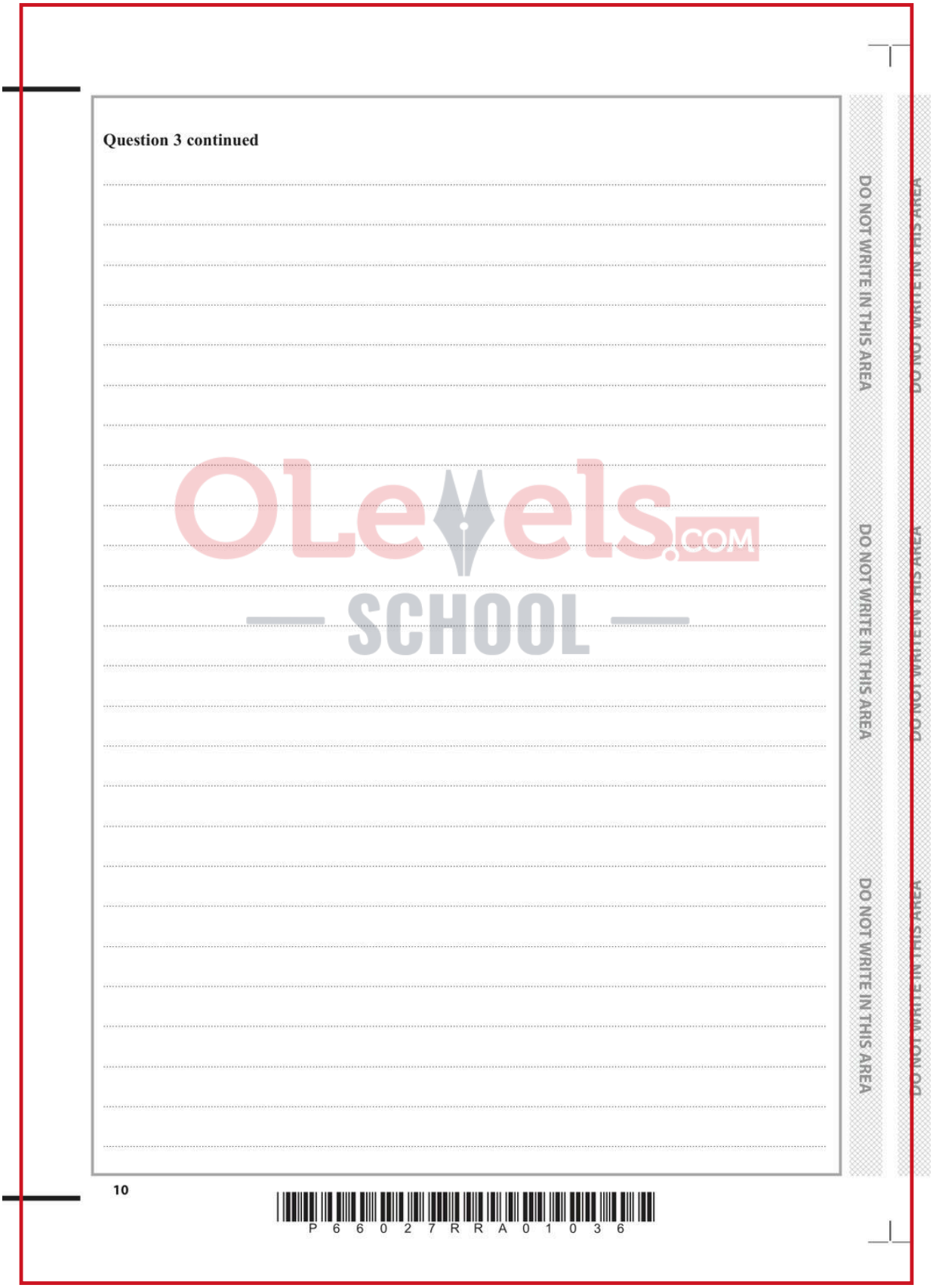
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 3 · part d · 1 mark

2021 NOV · Paper 02

Worked-example retry · Attempt part d only.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

10

P 6 6 0 2 7 R R A 0 1 0 3 6

1. Question 3 · part d · 1 mark

2021 NOV · Paper 02

Worked-example retry · Attempt part d only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 3 is 7 marks)

1. Question 3 · part d · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part b · 1 mark

2023 JAN · Paper 01

Further practice · Attempt part b only.

3 Curve C has equation $y = \dfrac{ax + 3}{1 - 2x}$ where $x \neq \dfrac{1}{2}$ and a is a constant.

The asymptote to C that is parallel to the x-axis has equation $y = 4$

(a) Find the value of a

(2)

(b) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

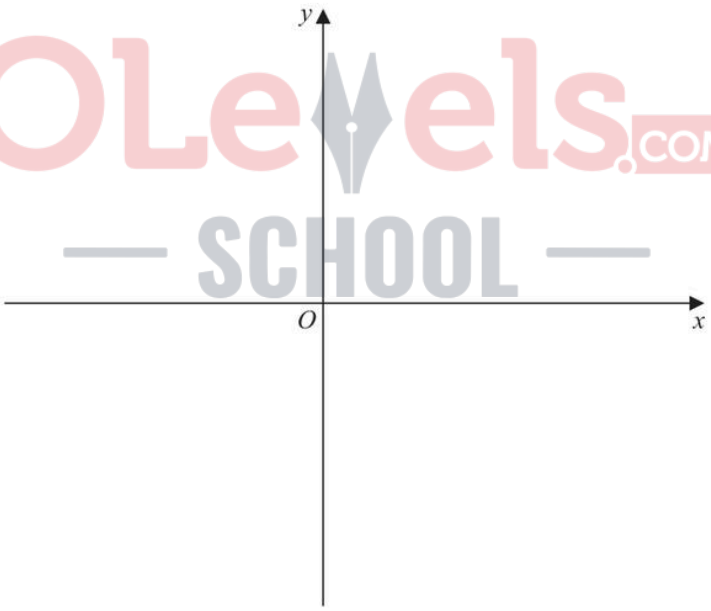
(c) Find the coordinates of the point where C crosses

(i) the x-axis, (ii) the y-axis.

(2)

(d) Using the axes below, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(4)



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 1 8 1 7 A 0 8 3 2

2. Question 3 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 10 · part a.i · 0 marks

2022 JAN · Paper 02

Further practice · Attempt part a.i only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

3. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part a.ii · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part a.ii only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

4. Question 10 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 10 · part a.i · 0 marks

2025 JUN · Paper 02

Further practice · Attempt part a.i only.

10 A curve C has equation

$$y = \frac{7-x}{2x+3} \qquad x \neq -\frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis

(ii) parallel to the x-axis

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

C passes through the point A with coordinates $\left(\frac{1}{2}, \frac{13}{8}\right)$

(d) Show that the gradient of C at A is $-\frac{17}{16}$

(3)

C also passes through the point P

Given that the tangent to C at P is parallel to the tangent to C at A

(e) find an equation of the tangent to C at P
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

5. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 10 · part a.ii · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part a.ii only.

10 A curve C has equation

$$y = \frac{7-x}{2x+3} \qquad x \neq -\frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis

(ii) parallel to the x-axis

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

C passes through the point A with coordinates $\left(\frac{1}{2}, \frac{13}{8}\right)$

(d) Show that the gradient of C at A is $-\frac{17}{16}$

(3)

C also passes through the point P

Given that the tangent to C at P is parallel to the tangent to C at A

(e) find an equation of the tangent to C at P
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(7)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

6. Question 10 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 3 · part b.i · 0 marks

2021 JUN · Paper 01

Further practice · Attempt part b.i only.

3 A curve C has equation $y = \dfrac{ax - 3}{x + 5}$ where a is a constant and $x \neq -5$

The gradient of C at the point on the curve where $x = 2$ is $\dfrac{18}{49}$

(a) Show that $a = 3$

(3)

Hence

(b) write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis,

(2)

(c) find the coordinates of the point where C crosses

(i) the x-axis,

(ii) the y-axis.

(2)

(d) Sketch the curve C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 3 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 3 · part b.ii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

3 A curve C has equation $y = \dfrac{ax - 3}{x + 5}$ where a is a constant and $x \neq -5$

The gradient of C at the point on the curve where $x = 2$ is $\dfrac{18}{49}$

(a) Show that $a = 3$

(3)

Hence

(b) write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis,

(2)

(c) find the coordinates of the point where C crosses

(i) the x-axis,

(ii) the y-axis.

(2)

(d) Sketch the curve C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 7 · part a · 1 mark

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

7

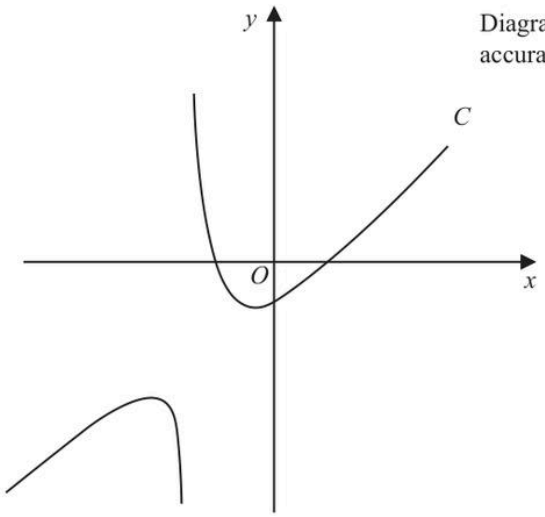


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a sketch of part of the curve C with equation

$$y = \frac{x^2 - 1}{4x + 5} \quad \text{where} \quad x \neq -\frac{5}{4}$$

(a) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

The line l is the normal to C at the point where $x = -1$

(b) Find an equation of l

(7)

The line l meets C again at the point D

(c) Find the coordinates of D

(6)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

14

P 7 4 2 8 5 R A 0 1 4 3 2

9. Question 7 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 3 · part a.i · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part a.i only.

3 Curve S has equation $y = 2 - \frac{5}{2x+1}$ where $x \neq -\frac{1}{2}$

(a) Write down an equation of the asymptote to S that is parallel to

(i) the y-axis (ii) the x-axis

(3)

(b) Find the coordinates of the point where S crosses

(i) the y-axis (ii) the x-axis

(2)

(c) Using the axes on the page opposite, sketch S, showing clearly the asymptotes and the coordinates of the points where S crosses the coordinate axes.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

10. Question 3 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 11 · part a.i · 0 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

11 The curve C has equation $y = \dfrac{3x - 2}{x + 1}$

(a) Write down an equation of the asymptote to C which is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

The straight line l has equation $y = mx + 4$

Given that there are **no** points of intersection between l and C,

(d) show algebraically that the range of possible values of m can be written as

$$a - 2\sqrt{b} < m < a + 2\sqrt{b}$$

where a and b are integers whose values need to be found.

(7)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

P 6 0 4 7 6 A 0 3 6 4 0

11. Question 11 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 11 · part a.ii · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

11 The curve C has equation $y = \dfrac{3x - 2}{x + 1}$

(a) Write down an equation of the asymptote to C which is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

The straight line l has equation $y = mx + 4$

Given that there are **no** points of intersection between l and C,

(d) show algebraically that the range of possible values of m can be written as

$$a - 2\sqrt{b} < m < a + 2\sqrt{b}$$

where a and b are integers whose values need to be found.

(7)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

12. Question 11 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 10 · part a.i · 0 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

10 A curve C has equation

$$y = \frac{7x - 2}{2x - 3} \qquad x \neq \frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis,

(ii) parallel to the x-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using calculus, show that at every point on the curve, the gradient of C is negative.

(4)

(d) Using the axes on the opposite page, sketch C.
Show clearly and label with their equation any asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The straight line l is the normal to C at the point A.
The x coordinate of A is positive and the gradient of l is 17
The line l also intersects C at the point B.

(e) Find the exact coordinates of B.

(7)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

13. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 10 · part a.ii · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a.ii only.

10 A curve C has equation

$$y = \frac{7x - 2}{2x - 3} \qquad x \neq \frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis,

(ii) parallel to the x-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using calculus, show that at every point on the curve, the gradient of C is negative.

(4)

(d) Using the axes on the opposite page, sketch C.
Show clearly and label with their equation any asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The straight line l is the normal to C at the point A.
The x coordinate of A is positive and the gradient of l is 17
The line l also intersects C at the point B.

(e) Find the exact coordinates of B.

(7)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

14. Question 10 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 9 · part a.i · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a.i only.

9 A curve C has equation $y = \dfrac{3 - 2x}{x + 6}$ where $x \neq -6$

(a) Write down an equation of the asymptote to C that is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Using the axes opposite, sketch the graph of C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

(d) Show that the gradient of the tangent to C is always negative.

(3)

A tangent to C has equation $y = -\dfrac{3}{5}x + k$ where $k > 0$

(e) Find the value of k

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

15. Question 9 · part a.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 10 · part b · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part b only.

10 The curve C with equation $y = \dfrac{6 - 3x}{x - 4}$ where $x \neq 4$, crosses the x-axis at the point P and the y-axis at the point Q

(a) Find the coordinates of

(i) P (ii) Q

(2)

(b) Write down an equation of the asymptote to C which is

(i) parallel to the y-axis (ii) parallel to the x-axis

(2)

(c) Sketch C showing clearly the asymptotes and the coordinates of the points P and Q

(3)

The line L is the normal to C at the point on C where $x = 2$

(d) Find an equation of L

(6)

The line L intersects C again at the point R

(e) Find the x coordinate of R

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

16. Question 10 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

17. Question 10 · part b.i · 0 marks

2024 NOV · Paper 02

Further practice · Attempt part b.i only.

10 A curve C has equation

$$y = \frac{5x-2}{3x+2} \qquad x \neq -\frac{2}{3}$$

(a) Find the coordinates of the point where C intersects the

(i) x-axis
(ii) y-axis

(2)

(b) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis
(ii) parallel to the y-axis

(2)

(c) Sketch C on the opposite page.
Show and label the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

Point A lies on C such that the gradient of C at A is parallel to the line with equation $4y - x = 7$

The normal to C at A intersects the x-axis at point D and the y-axis at point E
Given that the x coordinate of A is positive,

(d) find, in its simplified form, the exact length of line DE

(11)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

17. Question 10 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

18. Question 10 · part b.ii · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part b.ii only.

10 A curve C has equation

$$y = \frac{5x-2}{3x+2} \qquad x \neq -\frac{2}{3}$$

(a) Find the coordinates of the point where C intersects the

(i) x-axis
(ii) y-axis

(2)

(b) Write down an equation of the asymptote to C that is

(i) ~~parallel to the x-axis~~
(ii) parallel to the y-axis

(2)

(c) Sketch C on the opposite page.
Show and label the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

Point A lies on C such that the gradient of C at A is parallel to the line with equation $4y - x = 7$

The normal to C at A intersects the x-axis at point D and the y-axis at point E
Given that the x coordinate of A is positive,

(d) find, in its simplified form, the exact length of line DE

(11)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

P 7 3 5 8 7 A 0 2 8 3 2

18. Question 10 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

19. Question 10 · part b · 1 mark

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

(ii) Find the value of a

(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

(c) Find the coordinates of the point where C crosses the x-axis.

(1)

(d) Using calculus, show that at every point on C, the gradient is positive.

(4)

(e) Using the axes on the next page, sketch C
Clearly label the asymptotes and the coordinates where C crosses the coordinate axes.

(3)

The gradient of C is $\frac{13}{100}$ at the point E and at the point F

(f) Find the length of the line EF, giving your answer to 2 decimal places.

(6)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 6 5 0 9 A 0 2 6 3 2

19. Question 10 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Where a rational curve crosses the axes

Finding the coordinates where a curve y = (ax + b)/(cx + d) meets the x-axis and the y-axis.

16 question parts: 1 worked-example retry and 15 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-GRA-05

1. Question 3 · part c · 2 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

3

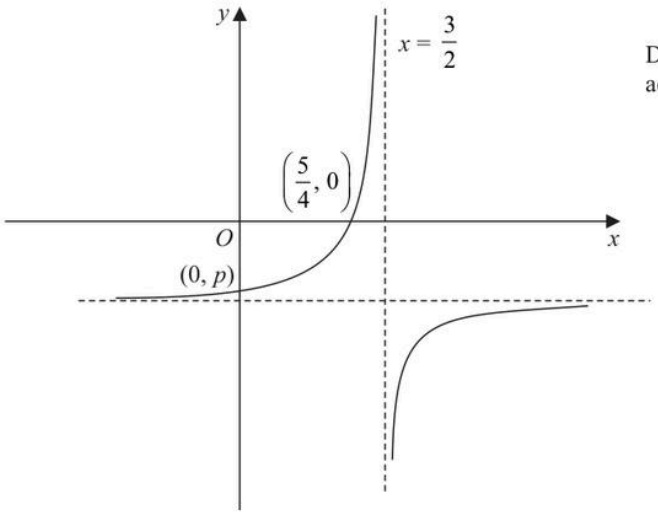


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a sketch of the curve with equation

$$y = \frac{a - bx}{cx - d} \qquad x \neq \frac{d}{c}$$

where a, c and d are prime numbers and b is an integer.

The asymptote to the curve that is parallel to the y-axis has equation $x = \frac{3}{2}$

(a) Write down the value of c and the value of d

(2)

The curve crosses the x-axis at the point $\left(\frac{5}{4}, 0\right)$

(b) Find the value of a and the value of b

(2)

The curve crosses the y-axis at the point $(0, p)$ where p is a rational number.

(c) Find the value of p

(2)

(d) Find an equation of the asymptote to the curve that is parallel to the x-axis.

(1)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

P 6 6 0 2 7 R R A 0 8 3 6

1. Question 3 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part c.i · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part c.i only.

3 Curve C has equation $y = \dfrac{ax + 3}{1 - 2x}$ where $x \neq \dfrac{1}{2}$ and a is a constant.

The asymptote to C that is parallel to the x-axis has equation $y = 4$

(a) Find the value of a

(2)

(b) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

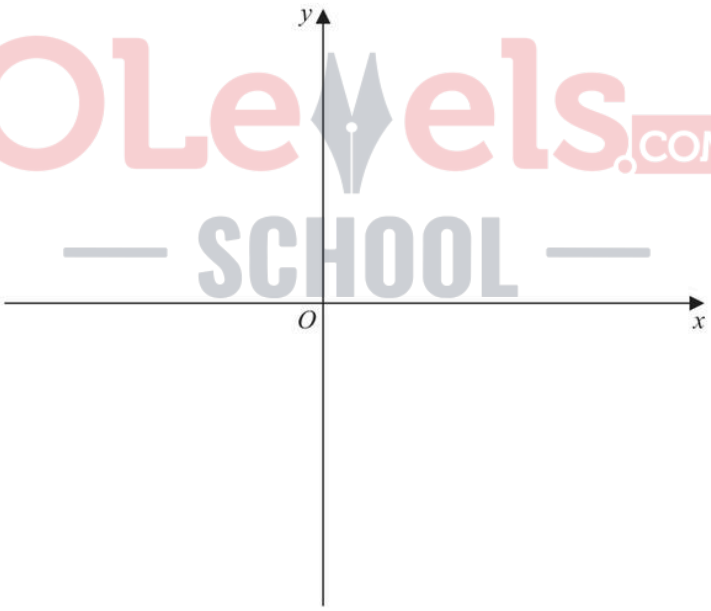
(c) Find the coordinates of the point where C crosses

(i) the x-axis, (ii) the y-axis.

(2)

(d) Using the axes below, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(4)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

8

P 7 1 8 1 7 A 0 8 3 2

2. Question 3 · part c.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 10 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

3. Question 10 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part b · 2 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

10 A curve C has equation

$$y = \frac{7-x}{2x+3} \qquad x \neq -\frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis

(ii) parallel to the x-axis

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

C passes through the point A with coordinates $\left(\frac{1}{2}, \frac{13}{8}\right)$

(d) Show that the gradient of C at A is $-\frac{17}{16}$

(3)

C also passes through the point P

Given that the tangent to C at P is parallel to the tangent to C at A

(e) find an equation of the tangent to C at P
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 10 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part c.i · 0 marks

2021 JUN · Paper 01

Further practice · Attempt part c.i only.

3 A curve C has equation $y = \dfrac{ax - 3}{x + 5}$ where a is a constant and $x \neq -5$

The gradient of C at the point on the curve where $x = 2$ is $\dfrac{18}{49}$

(a) Show that $a = 3$

(3)

Hence

(b) write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis,

(2)

(c) find the coordinates of the point where C crosses

(i) the x-axis,

(ii) the y-axis.

(2)

(d) Sketch the curve C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 3 · part c.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 3 · part c.ii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c.ii only.

3 A curve C has equation $y = \dfrac{ax - 3}{x + 5}$ where a is a constant and $x \neq -5$

The gradient of C at the point on the curve where $x = 2$ is $\dfrac{18}{49}$

(a) Show that $a = 3$

(3)

Hence

(b) write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis,

(2)

(c) find the coordinates of the point where C crosses

(i) the x-axis,

(ii) the y-axis.

(2)

(d) Sketch the curve C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Olevels.com SCHOOL

6. Question 3 · part c.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 3 · part b.i · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b.i only.

3 Curve S has equation $y = 2 - \frac{5}{2x+1}$ where $x \neq -\frac{1}{2}$

(a) Write down an equation of the asymptote to S that is parallel to

(i) the y-axis (ii) the x-axis

(3)

(b) Find the coordinates of the point where S crosses

(i) the y-axis (ii) the x-axis

(2)

(c) Using the axes on the page opposite, sketch S, showing clearly the asymptotes and the coordinates of the points where S crosses the coordinate axes.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 8 9 6 5 A 0 6 3 2

7. Question 3 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 11 · part b.i · 0 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b.i only.

11 The curve C has equation $y = \dfrac{3x - 2}{x + 1}$

(a) Write down an equation of the asymptote to C which is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

The straight line l has equation $y = mx + 4$

Given that there are **no** points of intersection between l and C,

(d) show algebraically that the range of possible values of m can be written as

$$a - 2\sqrt{b} < m < a + 2\sqrt{b}$$

where a and b are integers whose values need to be found.

(7)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

8. Question 11 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 11 · part b.ii · 2 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b.ii only.

11 The curve C has equation $y = \dfrac{3x - 2}{x + 1}$

(a) Write down an equation of the asymptote to C which is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

The straight line l has equation $y = mx + 4$

Given that there are **no** points of intersection between l and C,

(d) show algebraically that the range of possible values of m can be written as

$$a - 2\sqrt{b} < m < a + 2\sqrt{b}$$

where a and b are integers whose values need to be found.

(7)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

9. Question 11 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 10 · part b · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

10 A curve C has equation

$$y = \frac{7x - 2}{2x - 3} \qquad x \neq \frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis,

(ii) parallel to the x-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using calculus, show that at every point on the curve, the gradient of C is negative.

(4)

(d) Using the axes on the opposite page, sketch C.
Show clearly and label with their equation any asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The straight line l is the normal to C at the point A.
The x coordinate of A is positive and the gradient of l is 17
The line l also intersects C at the point B.

(e) Find the exact coordinates of B.

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

10. Question 10 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 9 · part b.i · 2 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b.i only.

9 A curve C has equation $y = \dfrac{3 - 2x}{x + 6}$ where $x \neq -6$

(a) Write down an equation of the asymptote to C that is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Using the axes opposite, sketch the graph of C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

(d) Show that the gradient of the tangent to C is always negative.

(3)

A tangent to C has equation $y = -\dfrac{3}{5}x + k$ where $k > 0$

(e) Find the value of k

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

11. Question 9 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 10 · part a · 2 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

10 The curve C with equation $y = \dfrac{6 - 3x}{x - 4}$ where $x \neq 4$, crosses the x-axis at the point P and the y-axis at the point Q

(a) Find the coordinates of

(i) P (ii) Q

(2)

(b) Write down an equation of the asymptote to C which is

(i) parallel to the y-axis (ii) parallel to the x-axis

(2)

(c) Sketch C showing clearly the asymptotes and the coordinates of the points P and Q

(3)

The line L is the normal to C at the point on C where $x = 2$

(d) Find an equation of L

(6)

The line L intersects C again at the point R

(e) Find the x coordinate of R

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

P 7 2 8 6 6 A 0 3 0 3 6

12. Question 10 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 10 · part a.i · 0 marks

2024 NOV · Paper 02

Further practice · Attempt part a.i only.

10 A curve C has equation

$$y = \frac{5x-2}{3x+2} \qquad x \neq -\frac{2}{3}$$

(a) Find the coordinates of the point where C intersects the

(i) x-axis
(ii) y-axis

(2)

(b) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis
(ii) parallel to the y-axis

(2)

(c) Sketch C on the opposite page.
Show and label the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

Point A lies on C such that the gradient of C at A is parallel to the line with equation $4y - x = 7$

The normal to C at A intersects the x-axis at point D and the y-axis at point E
Given that the x coordinate of A is positive,

(d) find, in its simplified form, the exact length of line DE

(11)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

13. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 10 · part a.ii · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part a.ii only.

10 A curve C has equation

$$y = \frac{5x-2}{3x+2} \qquad x \neq -\frac{2}{3}$$

(a) Find the coordinates of the point where C intersects the

(i) x-axis
(ii) y-axis

(2)

(b) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis
(ii) parallel to the y-axis

(2)

(c) Sketch C on the opposite page.
Show and label the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

Point A lies on C such that the gradient of C at A is parallel to the line with equation $4y - x = 7$

The normal to C at A intersects the x-axis at point D and the y-axis at point E
Given that the x coordinate of A is positive,

(d) find, in its simplified form, the exact length of line DE

(11)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

14. Question 10 · part a.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 10 · part c · 1 mark

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

(ii) Find the value of a

(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

(c) Find the coordinates of the point where C crosses the x-axis.

(1)

(d) Using calculus, show that at every point on C, the gradient is positive.

(4)

(e) Using the axes on the next page, sketch C
Clearly label the asymptotes and the coordinates where C crosses the coordinate axes.

(3)

The gradient of C is $\frac{13}{100}$ at the point E and at the point F

(f) Find the length of the line EF, giving your answer to 2 decimal places.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

15. Question 10 · part c · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 4 · part c · 2 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

4

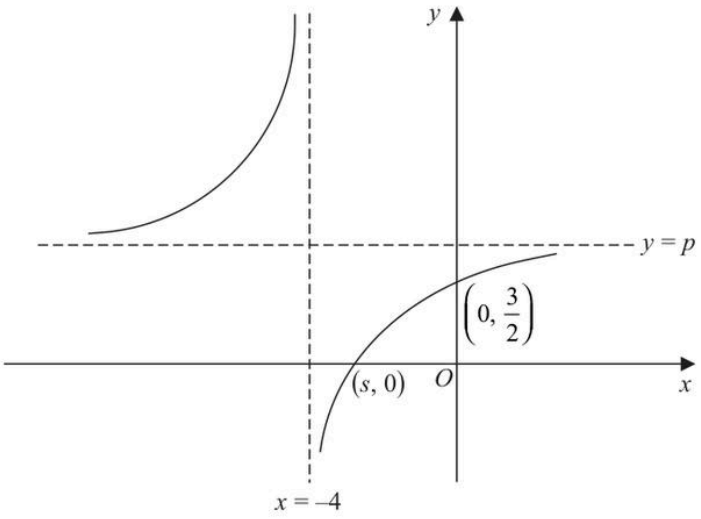


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows part of the curve C with equation

$$y = \frac{2x + q}{x + r} \qquad x \neq -r$$

where q and r are integers.

The asymptote to C that is parallel to the y-axis has equation $x = -4$
The asymptote to C that is parallel to the x-axis has equation $y = p$

(a) Write down

(i) the value of p

(ii) the value of r

(2)

Given that C crosses the y-axis at the point with coordinates $\left(0, \frac{3}{2}\right)$

(b) find the value of q

(2)

Given that C crosses the x-axis at the point with coordinates $(s, 0)$

(c) find the value of s

(2)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

P 7 1 8 2 0 A 0 8 3 6

16. Question 4 · part c · 2 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 4 is 6 marks)

16. Question 4 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Filling in a table of values

Working out the missing y-values in a table for a given curve (polynomial, reciprocal, exponential, logarithmic or trigonometric) to the stated number of decimal places.

15 question parts: 1 worked-example retry and 14 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-GRA-01

1. Question 7 · part a · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part a only.

7 (a) Complete the table of values for $y = \log_3(4-x)+3$ giving your answers to 2 decimal places.

x	0	0.5	1	1.5	2	2.5	3	3.5
y	4.26		4		3.63		3	2.37

(2)

(b) On the grid opposite, draw the graph of $y = \log_3(4-x)+3$ in the interval $0 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate,

to one decimal place, of the root of the equation $3^{2x-5} - (4-x)^3 = 0$ in the interval $0 \leqslant x \leqslant 3.5$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 2 8 6 7 A 0 1 4 3 2

1. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 8 · part a · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

8 (a) Complete the table of values for $y = 2 + \ln(2x + 1)$ giving your answers to 2 decimal places.

(2)

x	0	0.25	0.5	1	1.5	2	3
y	2			3.10	3.39	3.61	

(b) On the grid opposite, draw the graph of $y = 2 + \ln(2x + 1)$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $\ln(2x + 1) = 3x - 4$ in the interval $0 \leqslant x \leqslant 3$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $e^{(6-x)} - (2x + 1)^2 = 0$ in the interval $0 \leqslant x \leqslant 3$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

20

P 6 1 8 8 2 A 0 2 0 3 6

2. Question 8 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part a · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part a only.

7 (a) Complete the table of values for

$$y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$$

giving each value to 2 decimal places where appropriate.

x	–6	–5	–4	–3	–2	–1	0
y	4	3.59	3.26				2.5

(2)

(b) On the grid opposite, draw the graph of $y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$ for $-6 \leqslant x \leqslant 0$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_2(2x+2)^3 + x + 3 = 0 \text{ in the interval } -6 \leqslant x \leqslant 0$$

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

3. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part a · 2 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

4 (a) Complete the table of values for $y = 2x + 1 + \frac{2}{x^2}$

Give your answers to 2 decimal places where appropriate.

x	0.5	1	1.5	2	2.5	3	3.5
y		5			6.32		8.16

(2)

(b) On the grid opposite, draw the graph of $y = 2x + 1 + \frac{2}{x^2}$ for $0.5 \leqslant x \leqslant 3.5$

(2)

(c) Use your graph to obtain estimates, to 1 decimal place, of the roots of the equation

$$2x + \frac{2}{x^2} = 7 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(2)

(d) By drawing a suitable straight line on the grid, obtain estimates, to 1 decimal place, of the roots of the equation

$$\frac{3x}{2} + \frac{2}{x^2} = 5 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 4 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 8 · part a · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part a only.

8 (a) Given that x is measured in radians, complete the table of values for

$$y = \sin x - 3\cos 2x - 0.5$$

Give your answers to one decimal place.

x	0	0.5	0.8	1	1.6	2	2.5	3
y			0.3	1.6			–0.8	–3.2

(2)

(b) On the grid opposite, draw the graph of $y = \sin x - 3\cos 2x - 0.5$ for $0 \leqslant x \leqslant 3$

(2)

(c) Use a formula from page 2 to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

$$f(x) = 2\sin x + 12\sin^2 x - x - 5$$

(d) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $f(x) = 0$ in the interval $0 \leqslant x \leqslant 3$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 8 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 6 · part a · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 (a) Complete the table of values for

$$y = x - \frac{3}{x^2}$$

giving your answers to one decimal place where appropriate.

x	0.5	1	1.5	2	3	4	5	6
y	–11.5			1.3	2.7			5.9

(2)

(b) On the grid opposite, draw the graph of $y = x - \frac{3}{x^2}$ for $0.5 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of each of the two roots of the equation

$$2x^3 - 6x^2 + 3 = 0$$

in the interval $0.5 \leqslant x \leqslant 6$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

P 6 0 4 7 5 A 0 1 2 3 6

6. Question 6 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 4 · part a · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

4 (a) Complete the table of values for $y = \frac{x}{2} + 6e^{-2x} + 1$

giving your answers to one decimal place.

x	0	1	1.5	2	3	4	5	6
y	7		2.0			3.0		4.0

(2)

(b) On the grid opposite, draw the graph of $y = \frac{x}{2} + 6e^{-2x} + 1$ for $0 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on your graph, obtain estimates, to one decimal place, of the roots of the equation

$$2x + \ln(24 - 5x) = \ln 36$$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 4 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 8 · part a · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part a only.

8 (a) Complete the table of values for $y = 3^x - 2$ giving your answers to one decimal place.

x	-1	-0.5	0	0.25	0.5	0.75	1	1.5	2
y	-1.7		-1				1	3.2	7

(2)

(b) On the grid opposite, draw the graph of $y = 3^x - 2$ for $-1 \leqslant x \leqslant 2$

(2)

(c) Use your graph to obtain an estimate, to one decimal place, of the value of $\log_3 4.5$

Show clearly how you have used your graph.

(3)

(d) By drawing a straight line on your grid obtain an estimate, to one decimal place, of the root of the equation $\log_3(8-3x) - x = 0$ in the interval $-1 \leqslant x \leqslant 2$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 8 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 5 · part a · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

5 (a) Complete the table of values for $y = e^{3x-2}$ giving your answers to 2 decimal places.

x	0	0.25	0.5	0.75	1
y	0.14				2.72

(2)

(b) On the grid opposite, draw the graph of $y = e^{3x-2}$ for $0 \leqslant x \leqslant 1$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $3x = 2 + \ln(3 - x)$

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 1 6 4 2 A 0 1 2 3 6

9. Question 5 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 8 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

8 (a) Complete the table of values for $y = 2x + \frac{3}{x^2} - 3$ giving your answers to 2 decimal places where appropriate.

x	0.5	0.75	1	1.5	2	3	4	5
y	10		2				5.19	7.12

(2)

(b) On the grid opposite, draw the graph of $y = 2x + \frac{3}{x^2} - 3$ for $0.5 \leqslant x \leqslant 5$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $4x^3 - 10x^2 + 3 = 0$ for $0.5 \leqslant x \leqslant 5$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 8 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 4 · part a · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part a only.

4 (a) Complete the table of values for $y = \log_{10}(6x-1) - x$ giving your answers to 2 decimal places.

x	0.25	0.5	1	1.5	2	2.5	3
y		–0.20	–0.30	–0.60			

(2)

(b) On the grid opposite, draw the graph of $y = \log_{10}(6x-1) - x$ for $0.25 \leqslant x \leqslant 3$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$10^{\frac{3x-4}{2}} = 6x - 1 \text{ in the interval } 0.25 \leqslant x \leqslant 3$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8

P 7 3 5 8 7 A 0 8 3 2

11. Question 4 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 9 · part a · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

9 (a) Complete the table of values for

$$y = \frac{x^3 + 4}{5 - x}$$

giving your answers to 2 decimal places.

x	−2	−1.5	−1	−0.5	0	0.5	1	1.5	2
y	−0.57	0.10	0.5		0.8		1.25		4

(2)

(b) On the grid opposite, draw the graph of

$$y = \frac{x^3 + 4}{5 - x} \quad \text{for } -2 \leqslant x \leqslant 2$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $x^3 - x^2 + 8x - 11 = 0$ in the interval $-2 \leqslant x \leqslant 2$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

12. Question 9 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 7 · part a · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

7 (a) Complete the table of values for $y = 3\log_3\left(x^2 - x\right) - 2x$ giving your answers to 2 decimal places.

x	1.3	1.5	2	2.5	3	3.5
y		−3.79	−2.11			

(2)

(b) On the grid opposite, draw the graph of $y = 3\log_3\left(x^2 - x\right) - 2x$ for $1.3 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$3^{\frac{4}{3}x} = 9\left(x^2 - x\right)^3 \quad \text{in the interval } 1.3 \leqslant x \leqslant 3.5$$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

13. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 7 · part a · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

7 (a) Complete the table of values for $y = 3^{\frac{x}{4}} + 2$

Give your answers to 2 decimal places where appropriate.

(2)

x	0	1	2	3	4	5
y	3	3.32				5.95

(b) On the grid opposite, draw the graph of

$$y = 3^{\frac{x}{4}} + 2 \quad \text{for} \quad 0 \leqslant x \leqslant 5$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_3(6 - 2x)^4 - x = 0$$

in the interval $0 \leqslant x \leqslant 5$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

14. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 6 · part a · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part a only.

6 (a) Complete the table of values for $y = 1 + 3e^{-x}$ giving your answers to 2 decimal places where appropriate.

(2)

x	0	0.25	0.5	1	1.5	2	3
y		3.34	2.82		1.67		1.15

(b) On the grid opposite, draw the graph of $y = 1 + 3e^{-x}$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$x = e^{-x} \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\ln(x-1)^3 = -3x \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(4)

OLevels.COM — SCHOOL —

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

15. Question 6 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Drawing the graph from your table

Plotting the points from a completed table on the printed grid and joining them with a smooth curve over the given interval.

15 question parts: 1 worked-example retry and 14 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-GRA-02

1. Question 7 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

7 (a) Complete the table of values for $y = \log_3(4-x)+3$ giving your answers to 2 decimal places.

x	0	0.5	1	1.5	2	2.5	3	3.5
y	4.26		4		3.63		3	2.37

(2)

(b) On the grid opposite, draw the graph of $y = \log_3(4-x)+3$ in the interval $0 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate,

to one decimal place, of the root of the equation $3^{2x-5} - (4-x)^3 = 0$ in the interval $0 \leqslant x \leqslant 3.5$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 2 8 6 7 A 0 1 4 3 2

1. Question 7 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 8 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

8 (a) Complete the table of values for $y = 2 + \ln(2x + 1)$ giving your answers to 2 decimal places.

(2)

x	0	0.25	0.5	1	1.5	2	3
y	2			3.10	3.39	3.61	

(b) On the grid opposite, draw the graph of $y = 2 + \ln(2x + 1)$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $\ln(2x + 1) = 3x - 4$ in the interval $0 \leqslant x \leqslant 3$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $e^{(6-x)} - (2x + 1)^2 = 0$ in the interval $0 \leqslant x \leqslant 3$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

20

P 6 1 8 8 2 A 0 2 0 3 6

2. Question 8 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part b · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

7 (a) Complete the table of values for

$$y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$$

giving each value to 2 decimal places where appropriate.

x	–6	–5	–4	–3	–2	–1	0
y	4	3.59	3.26				2.5

(2)

(b) On the grid opposite, draw the graph of $y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$ for $-6 \leqslant x \leqslant 0$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_2(2x+2)^3 + x + 3 = 0 \text{ in the interval } -6 \leqslant x \leqslant 0$$

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

3. Question 7 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part b · 2 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

4 (a) Complete the table of values for $y = 2x + 1 + \frac{2}{x^2}$

Give your answers to 2 decimal places where appropriate.

x	0.5	1	1.5	2	2.5	3	3.5
y		5			6.32		8.16

(2)

(b) On the grid opposite, draw the graph of $y = 2x + 1 + \frac{2}{x^2}$ for $0.5 \leqslant x \leqslant 3.5$

(2)

(c) Use your graph to obtain estimates, to 1 decimal place, of the roots of the equation

$$2x + \frac{2}{x^2} = 7 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(2)

(d) By drawing a suitable straight line on the grid, obtain estimates, to 1 decimal place, of the roots of the equation

$$\frac{3x}{2} + \frac{2}{x^2} = 5 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 4 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 8 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

8 (a) Given that x is measured in radians, complete the table of values for

$$y = \sin x - 3\cos 2x - 0.5$$

Give your answers to one decimal place.

x	0	0.5	0.8	1	1.6	2	2.5	3
y			0.3	1.6			–0.8	–3.2

(2)

(b) On the grid opposite, draw the graph of $y = \sin x - 3\cos 2x - 0.5$ for $0 \leqslant x \leqslant 3$

(2)

(c) Use a formula from page 2 to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

$$f(x) = 2\sin x + 12\sin^2 x - x - 5$$

(d) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $f(x) = 0$ in the interval $0 \leqslant x \leqslant 3$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

5. Question 8 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 6 · part b · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 (a) Complete the table of values for

$$y = x - \frac{3}{x^2}$$

giving your answers to one decimal place where appropriate.

x	0.5	1	1.5	2	3	4	5	6
y	–11.5			1.3	2.7			5.9

(2)

(b) On the grid opposite, draw the graph of $y = x - \frac{3}{x^2}$ for $0.5 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of each of the two roots of the equation

$$2x^3 - 6x^2 + 3 = 0$$

in the interval $0.5 \leqslant x \leqslant 6$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

P 6 0 4 7 5 A 0 1 2 3 6

6. Question 6 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 5 · part b · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 (a) Complete the table of values for $y = e^{3x-2}$ giving your answers to 2 decimal places.

x	0	0.25	0.5	0.75	1
y	0.14				2.72

(2)

(b) On the grid opposite, draw the graph of $y = e^{3x-2}$ for $0 \leqslant x \leqslant 1$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $3x = 2 + \ln(3 - x)$

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 5 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 8 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

8 (a) Complete the table of values for $y = 2x + \frac{3}{x^2} - 3$ giving your answers to 2 decimal places where appropriate.

x	0.5	0.75	1	1.5	2	3	4	5
y	10		2				5.19	7.12

(2)

(b) On the grid opposite, draw the graph of $y = 2x + \frac{3}{x^2} - 3$ for $0.5 \leqslant x \leqslant 5$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $4x^3 - 10x^2 + 3 = 0$ for $0.5 \leqslant x \leqslant 5$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 8 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 4 · part b · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

4 (a) Complete the table of values for $y = \log_{10}(6x-1) - x$ giving your answers to 2 decimal places.

x	0.25	0.5	1	1.5	2	2.5	3
y		–0.20	–0.30	–0.60			

(2)

(b) On the grid opposite, draw the graph of $y = \log_{10}(6x-1) - x$ for $0.25 \leqslant x \leqslant 3$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$10^{\frac{3x-4}{2}} = 6x - 1 \quad \text{in the interval } 0.25 \leqslant x \leqslant 3$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 3 5 8 7 A 0 8 3 2

9. Question 4 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 9 · part b · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

9 (a) Complete the table of values for

$$y = \frac{x^3 + 4}{5 - x}$$

giving your answers to 2 decimal places.

x	−2	−1.5	−1	−0.5	0	0.5	1	1.5	2
y	−0.57	0.10	0.5		0.8		1.25		4

(2)

(b) On the grid opposite, draw the graph of

$$y = \frac{x^3 + 4}{5 - x} \quad \text{for } -2 \leqslant x \leqslant 2$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $x^3 - x^2 + 8x - 11 = 0$ in the interval $-2 \leqslant x \leqslant 2$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 9 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 7 · part b · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 (a) Complete the table of values for $y = 3\log_3\left(x^2 - x\right) - 2x$ giving your answers to 2 decimal places.

x	1.3	1.5	2	2.5	3	3.5
y		−3.79	−2.11			

(2)

(b) On the grid opposite, draw the graph of $y = 3\log_3\left(x^2 - x\right) - 2x$ for $1.3 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$3^{\frac{4}{3}x} = 9\left(x^2 - x\right)^3 \quad \text{in the interval } 1.3 \leqslant x \leqslant 3.5$$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 6 5 0 7 A 0 1 4 3 2

11. Question 7 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 4 · part b · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

4 (a) Complete the table of values for $y = \frac{x}{2} + 6e^{-2x} + 1$

giving your answers to one decimal place.

x	0	1	1.5	2	3	4	5	6
y	7		2.0			3.0		4.0

(2)

(b) On the grid opposite, draw the graph of $y = \frac{x}{2} + 6e^{-2x} + 1$ for $0 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on your graph, obtain estimates, to one decimal place, of the roots of the equation

$$2x + \ln(24 - 5x) = \ln 36$$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

12. Question 4 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 8 · part b · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

8 (a) Complete the table of values for $y = 3^x - 2$ giving your answers to one decimal place.

x	-1	-0.5	0	0.25	0.5	0.75	1	1.5	2
y	-1.7		-1				1	3.2	7

(2)

(b) On the grid opposite, draw the graph of $y = 3^x - 2$ for $-1 \leqslant x \leqslant 2$

(2)

(c) Use your graph to obtain an estimate, to one decimal place, of the value of $\log_3 4.5$

Show clearly how you have used your graph.

(3)

(d) By drawing a straight line on your grid obtain an estimate, to one decimal place, of the root of the equation $\log_3(8-3x) - x = 0$ in the interval $-1 \leqslant x \leqslant 2$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

13. Question 8 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 7 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

7 (a) Complete the table of values for $y = 3^{\frac{x}{4}} + 2$

Give your answers to 2 decimal places where appropriate.

(2)

x	0	1	2	3	4	5
y	3	3.32				5.95

(b) On the grid opposite, draw the graph of

$$y = 3^{\frac{x}{4}} + 2 \quad \text{for} \quad 0 \leqslant x \leqslant 5$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_3(6 - 2x)^4 - x = 0$$

in the interval $0 \leqslant x \leqslant 5$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

14. Question 7 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 6 · part b · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

6 (a) Complete the table of values for $y = 1 + 3e^{-x}$ giving your answers to 2 decimal places where appropriate.

(2)

x	0	0.25	0.5	1	1.5	2	3
y		3.34	2.82		1.67		1.15

(b) On the grid opposite, draw the graph of $y = 1 + 3e^{-x}$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$x = e^{-x} \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\ln(x-1)^3 = -3x \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(4)

OLevels.COM — SCHOOL —

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

15. Question 6 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Sketching a rational curve with its asymptotes

Sketching the two branches of y = (ax + b)/(cx + d) with both asymptotes drawn and labelled and the axis intercepts marked.

17 question parts: 1 worked-example retry and 16 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-GRA-06

1. Question 9 · part b · 5 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part b only.

9 A curve C has equation

$$y = \frac{qx - 2}{x - p} \qquad x \neq p$$

The curve crosses the y-axis at the point A.

The line l with equation $y = x + 2$ is the normal to C at A.

(a) (i) Show that $p = 1$

(ii) Find the value of q.

(7)

(b) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(c) Find the x coordinate of D.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 9 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part d · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.

3 Curve C has equation $y = \dfrac{ax + 3}{1 - 2x}$ where $x \neq \dfrac{1}{2}$ and a is a constant.

The asymptote to C that is parallel to the x-axis has equation $y = 4$

(a) Find the value of a

(2)

(b) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

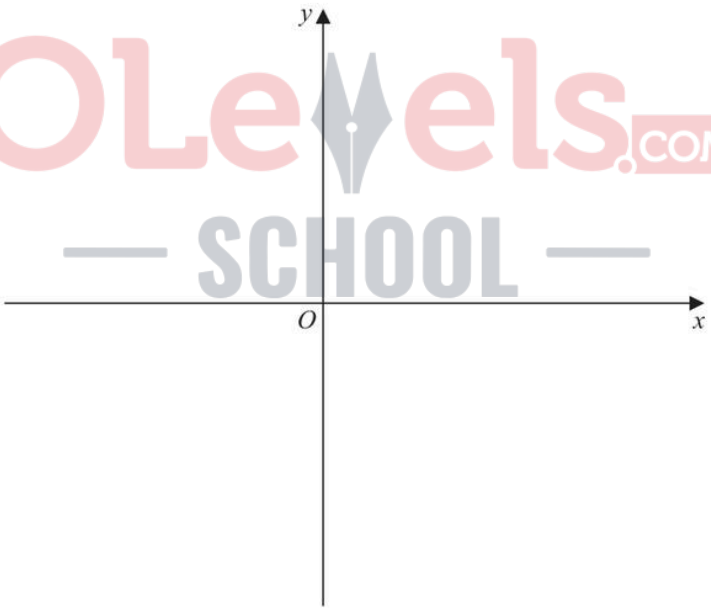
(c) Find the coordinates of the point where C crosses

(i) the x-axis, (ii) the y-axis.

(2)

(d) Using the axes below, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(4)



OLevels.COM — SCHOOL —

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 1 8 1 7 A 0 8 3 2

2. Question 3 · part d · 4 marks

2023 JAN · Paper 01

Further practice · Attempt part d only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 3 is 9 marks)

2. Question 3 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 10 · part e · 3 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part e only.

10 The curve C has equation

$$y = \frac{5x-2}{ax+b} \qquad x \neq -\frac{b}{a}$$

The asymptote to C that is parallel to the y-axis has equation $x = -\frac{4}{3}$

C crosses the y-axis at the point with coordinates $\left(0, -\frac{1}{4}\right)$

(a) (i) Show that $b = 8$

(ii) Find the value of a

(3)

(b) Write down the equation of the asymptote to C that is parallel to the x-axis.

(1)

(c) Find the coordinates of the point where C crosses the x-axis.

(1)

(d) Using calculus, show that at every point on C, the gradient is positive.

(4)

(e) Using the axes on the next page, sketch C
Clearly label the asymptotes and the coordinates where C crosses the coordinate axes.

(3)

The gradient of C is $\frac{13}{100}$ at the point E and at the point F

(f) Find the length of the line EF, giving your answer to 2 decimal places.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 6 5 0 9 A 0 2 6 3 2

3. Question 10 · part e · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part c · 3 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

10 The curve C has equation $y = \dfrac{2x-1}{x+4}$ $\quad x \neq -4$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The line with equation $y = x + k_1$ is the tangent to C at the point P

The line with equation $y = x + k_2$ is the tangent to C at the point Q

Given that the x coordinate of P is greater than the x coordinate of Q

(d) using calculus, find the coordinates of

(i) P

(ii) Q

(8)

(e) Hence find the value of

(i) k_1

(ii) k_2

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

4. Question 10 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 10 · part b · 5 marks

2024 JUN · Paper 01

Further practice · Attempt part b only.

10 The curve C has equation $y = \dfrac{ax - 5}{b - x}$ where a and b are integers and $x \neq b$

One intersection of C with the coordinate axes is at the point with coordinates $\left(\frac{5}{4}, 0\right)$

The asymptote parallel to the y-axis has equation $x = 3$

(a) Find the value of a and the value of b

(2)

(b) Sketch C, showing clearly the asymptotes with their equations and the coordinates of the points of intersection with the coordinate axes.

(5)

The straight line l with equation $4y - 7x = k$ has no points of intersection with C

(c) Show, using algebra, that the range of possible values of k can be written as

$$m < k < n$$

where m and n are integers to be found.

(9)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

5. Question 10 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 10 · part c · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

10 A curve C has equation

$$y = \frac{7-x}{2x+3} \qquad x \neq -\frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis

(ii) parallel to the x-axis

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

C passes through the point A with coordinates $\left(\frac{1}{2}, \frac{13}{8}\right)$

(d) Show that the gradient of C at A is $-\frac{17}{16}$

(3)

C also passes through the point P

Given that the tangent to C at P is parallel to the tangent to C at A

(e) find an equation of the tangent to C at P
Give your answer in the form $ax + by + c = 0$ where a, b and c are integers.

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 10 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 3 · part d · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

3 A curve C has equation $y = \dfrac{ax - 3}{x + 5}$ where a is a constant and $x \neq -5$

The gradient of C at the point on the curve where $x = 2$ is $\dfrac{18}{49}$

(a) Show that $a = 3$

(3)

Hence

(b) write down an equation of the asymptote to C that is

(i) parallel to the x-axis,

(ii) parallel to the y-axis,

(2)

(c) find the coordinates of the point where C crosses

(i) the x-axis,

(ii) the y-axis.

(2)

(d) Sketch the curve C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Olevels.com SCHOOL

7. Question 3 · part d · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 3 · part d · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 3 · part d · 3 marks

2021 JUN · Paper 01

Further practice · Attempt part d only.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

(Total for Question 3 is 10 marks)

7. Question 3 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 3 · part c · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

3 Curve S has equation $y = 2 - \frac{5}{2x+1}$ where $x \neq -\frac{1}{2}$

(a) Write down an equation of the asymptote to S that is parallel to

(i) the y-axis (ii) the x-axis

(3)

(b) Find the coordinates of the point where S crosses

(i) the y-axis (ii) the x-axis

(2)

(c) Using the axes on the page opposite, sketch S, showing clearly the asymptotes and the coordinates of the points where S crosses the coordinate axes.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

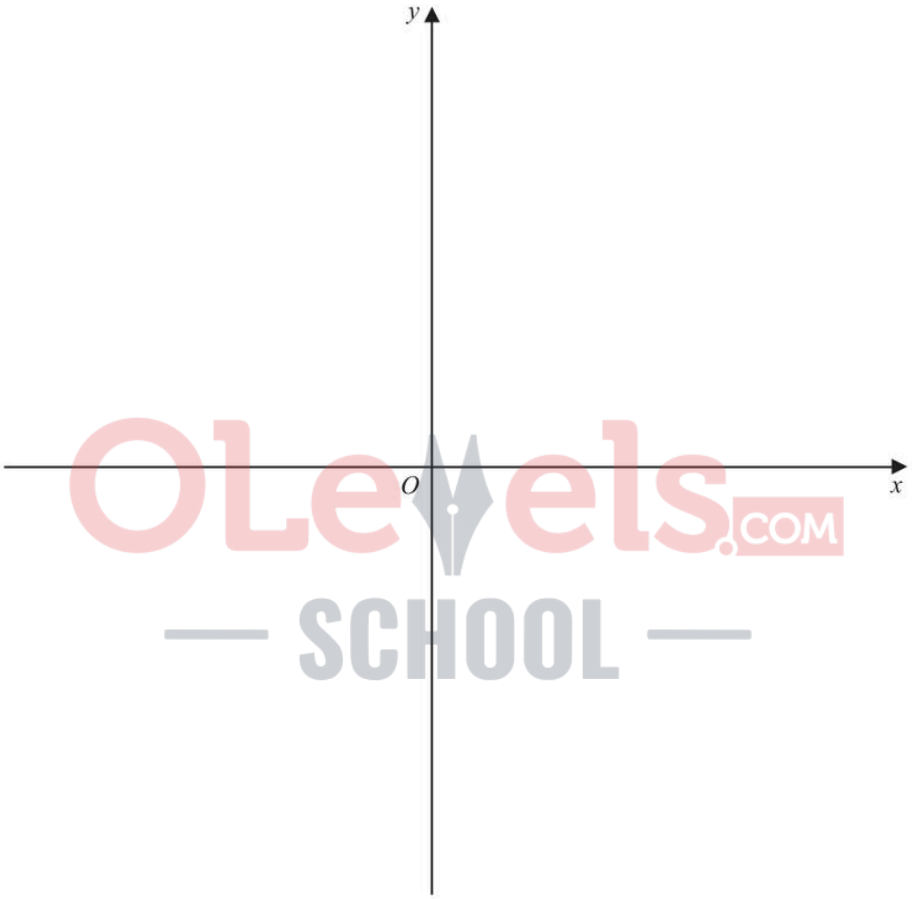
P 7 8 9 6 5 A 0 6 3 2

8. Question 3 · part c · 3 marks

2025 NOV · Paper 02

Further practice · Attempt part c only.

Question 3 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 3 is 8 marks)

OLevels.COM — SCHOOL —

8. Question 3 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 11 · part c · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

11 The curve C has equation $y = \dfrac{3x - 2}{x + 1}$

(a) Write down an equation of the asymptote to C which is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

The straight line l has equation $y = mx + 4$

Given that there are **no** points of intersection between l and C,

(d) show algebraically that the range of possible values of m can be written as

$$a - 2\sqrt{b} < m < a + 2\sqrt{b}$$

where a and b are integers whose values need to be found.

(7)

OLevels.COM — SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

9. Question 11 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 10 · part d · 3 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part d only.

10 A curve C has equation

$$y = \frac{7x - 2}{2x - 3} \qquad x \neq \frac{3}{2}$$

(a) Write down an equation of the asymptote to C that is

(i) parallel to the y-axis,

(ii) parallel to the x-axis.

(2)

(b) Find the coordinates of the points of intersection of C with the coordinate axes.

(2)

(c) Using calculus, show that at every point on the curve, the gradient of C is negative.

(4)

(d) Using the axes on the opposite page, sketch C.
Show clearly and label with their equation any asymptotes and the coordinates of the points of intersection of C with the coordinate axes.

(3)

The straight line l is the normal to C at the point A.
The x coordinate of A is positive and the gradient of l is 17
The line l also intersects C at the point B.

(e) Find the exact coordinates of B.

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

10. Question 10 · part d · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 9 · part c · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

9 A curve C has equation $y = \dfrac{3 - 2x}{x + 6}$ where $x \neq -6$

(a) Write down an equation of the asymptote to C that is parallel to the

(i) x-axis (ii) y-axis

(2)

(b) Find the coordinates of the point where C crosses the

(i) x-axis (ii) y-axis

(2)

(c) Using the axes opposite, sketch the graph of C, showing clearly its asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

(d) Show that the gradient of the tangent to C is always negative.

(3)

A tangent to C has equation $y = -\dfrac{3}{5}x + k$ where $k > 0$

(e) Find the value of k

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 9 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 11 · part c · 5 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

11 A curve C has equation

$$y = \frac{(2a-1)x+1}{ax-6} \qquad \text{where } a \text{ is a constant and } x \neq \frac{6}{a}$$

(a) Find $\frac{dy}{dx}$

(3)

The curve crosses the y-axis at the point A.

The normal to C at the point A is the line l with equation $66y - 72x + 11 = 0$

Show that

(b) (i) $a = 3$

(4)

(ii) the equation of C is $y = \frac{5x+1}{3x-6}$ where $x \neq 2$

(1)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes with their equations and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(d) Find the x coordinate of D.
Give your answer as an improper fraction.

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

12. Question 11 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 10 · part c · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part c only.

10 The curve C with equation $y = \dfrac{6 - 3x}{x - 4}$ where $x \neq 4$, crosses the x-axis at the point P and the y-axis at the point Q

(a) Find the coordinates of

(i) P (ii) Q

(2)

(b) Write down an equation of the asymptote to C which is

(i) parallel to the y-axis (ii) parallel to the x-axis

(2)

(c) Sketch C showing clearly the asymptotes and the coordinates of the points P and Q

(3)

The line L is the normal to C at the point on C where $x = 2$

(d) Find an equation of L

(6)

The line L intersects C again at the point R

(e) Find the x coordinate of R

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
OLevels.com SCHOOL

P 7 2 8 6 6 A 0 3 0 3 6

13. Question 10 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 10 · part c · 3 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

10 A curve C has equation

$$y = \frac{5x-2}{3x+2} \qquad x \neq -\frac{2}{3}$$

(a) Find the coordinates of the point where C intersects the

(i) x-axis
(ii) y-axis

(2)

(b) Write down an equation of the asymptote to C that is

(i) parallel to the x-axis
(ii) parallel to the y-axis

(2)

(c) Sketch C on the opposite page.
Show and label the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(3)

Point A lies on C such that the gradient of C at A is parallel to the line with equation $4y - x = 7$

The normal to C at A intersects the x-axis at point D and the y-axis at point E
Given that the x coordinate of A is positive,

(d) find, in its simplified form, the exact length of line DE

(11)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

14. Question 10 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 9 · part d.i · 0 marks

2020 NOV · Paper 02

Further practice · Attempt part d.i only.

9 A curve C has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of C in the form $ax^2 + (by - 4)x + (y - c) = 0$, where a, b and c are integers whose values are to be found.

(3)

(b) Hence show that x is real when $y \leqslant 2$ and when $y \geqslant 3$

(4)

(c) Find the coordinates of the stationary points on C.

(6)

(d) Sketch C showing clearly

(i) the exact coordinates of the points where C crosses the x-axis,

(ii) the asymptote to C that is parallel to the y-axis,

(iii) the coordinates of the stationary points.

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

15. Question 9 · part d.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 9 · part d.ii · 0 marks

2020 NOV · Paper 02

Further practice · Attempt part d.ii only.

9 A curve C has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of C in the form $ax^2 + (by - 4)x + (y - c) = 0$, where a, b and c are integers whose values are to be found.

(3)

(b) Hence show that x is real when $y \leqslant 2$ and when $y \geqslant 3$

(4)

(c) Find the coordinates of the stationary points on C.

(6)

(d) Sketch C showing clearly

(i) the exact coordinates of the points where C crosses the x-axis,

(ii) the asymptote to C that is parallel to the y-axis,

(iii) the coordinates of the stationary points.

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

16. Question 9 · part d.ii · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

17. Question 9 · part d.iii · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part d.iii only.

9 A curve C has equation

$$y = \frac{2 + 4x - x^2}{2x + 1} \quad x \neq -\frac{1}{2}$$

(a) Write the equation of C in the form $ax^2 + (by - 4)x + (y - c) = 0$, where a, b and c are integers whose values are to be found.

(3)

(b) Hence show that x is real when $y \leqslant 2$ and when $y \geqslant 3$

(4)

(c) Find the coordinates of the stationary points on C.

(6)

(d) Sketch C showing clearly

(i) the exact coordinates of the points where C crosses the x-axis,

(ii) the asymptote to C that is parallel to the y-axis,

(iii) the coordinates of the stationary points.

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

17. Question 9 · part d.iii · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part d.iii only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

17. Question 9 · part d.iii · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part d.iii only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

17. Question 9 · part d.iii · 5 marks

2020 NOV · Paper 02

Further practice · Attempt part d.iii only.



17. Question 9 · part d.iii · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Sketching curves and where they cross the axes

Key points and freehand sketches of non-rational curves: parabolas, cubics, exponential and log curves.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-GRA-03

1. Question 3 · part e · 2 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part e only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 1 8 1 9 A 0 6 3 6

1. Question 3 · part e · 2 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part e only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 3 · part e · 2 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part e only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 3 · part e · 2 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part e only.



1. Question 3 · part e · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

2 (a) Express $x^2 + 4x - 8$ in the form $(x + a)^2 + b$ where a and b are constants whose values are to be found.

(2)

(b) Use algebra to solve the simultaneous equations

$$y = x^2 + 4x - 8$$
$$y = 2x + 7$$

(5)

The curve C has equation $y = x^2 + 4x - 8$
The straight line L has equation $y = 2x + 7$

Using the same axes and the results of part (a) and part (b),

(c) sketch C and L, showing clearly the coordinates of the turning point of C and the coordinates of the points of intersection of C and L.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 2 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 2 continued

OLevels.com SCHOOL

2. Question 2 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8

P 6 2 2 8 3 A 0 8 4 0

2. Question 2 · part c · 4 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.



Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 2 is 11 marks)

2. Question 2 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 11 · part a · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

11

Diagram **NOT** accurately drawn

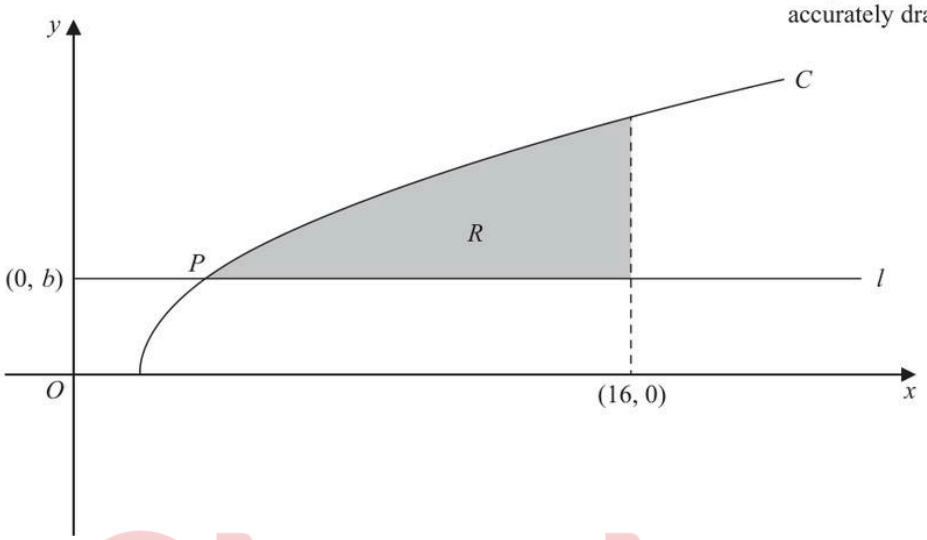


Figure 2

Figure 2 shows part of the curve C with equation $y = \sqrt{x - 2}$
Figure 2 also shows the straight line l with equation $y = b$ for $x > 0$ where $b > 0$

Given that C and l intersect at the point P with coordinates (a, b), where $2 < a < 16$

(a) show that $b^2 = a - 2$

(2)

The finite region R bounded by C, the straight line with equation $x = 16$ and l, shown shaded in Figure 2, is rotated through 360° about the x-axis to form a solid S.

Given that the volume of the solid formed is 50π

(b) use algebraic integration to find the value of a and the value of b.

(9)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 6 0 4 7 5 A 0 3 2 3 6

3. Question 11 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 11 · part a · 1 mark

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

11

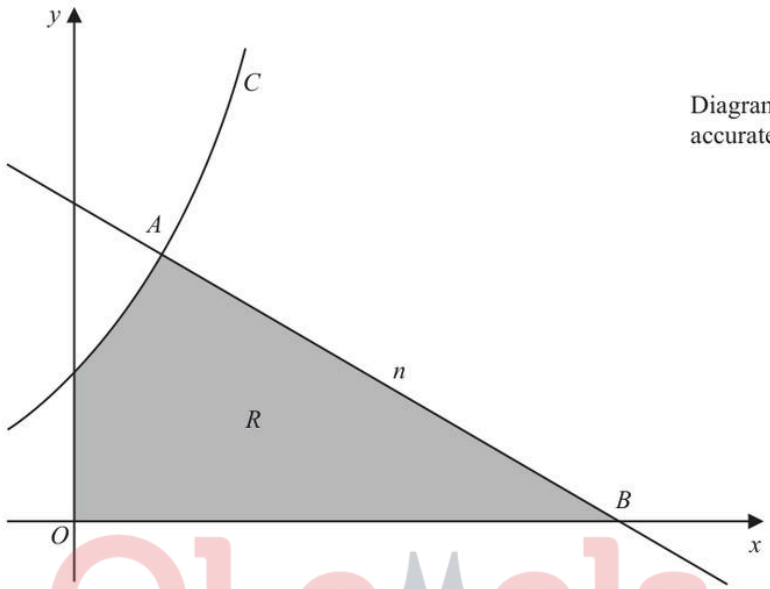


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of a curve C with equation $y = 3e^{\frac{x}{3}}$
The point A with coordinates $(2, p)$ lies on C

(a) Write down the exact value of p

(1)

The straight line n, shown on Figure 4, is the normal to C at A and crosses the x-axis at point B

The region R, shown shaded in Figure 4, is bounded by C, n, the x-axis and the y-axis.

(b) Use algebraic integration to find the area of R

Give your answer in the form $\frac{W}{V}e^2 + We^{\frac{2}{3}} - W$

where W and V are integers to be found.

(11)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 6 5 0 9 A 0 3 0 3 2

4. Question 11 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 11 · part a.i · 0 marks

2023 JAN · Paper 02

Further practice · Attempt part a.i only.

11

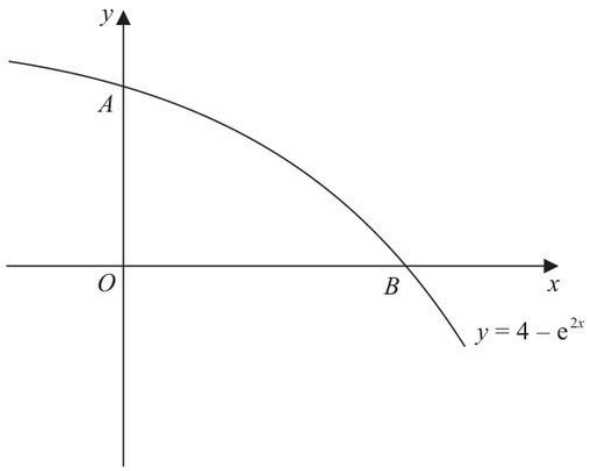


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows part of the curve C with equation $y = 4 - e^{2x}$
The curve C crosses the y-axis at the point A and the x-axis at the point B.

(a) (i) Write down the y coordinate of point A.

(ii) Show that the x coordinate of B is $x = \ln 2$

(3)

The line l is the normal to C at the point B.

(b) Find an equation for l, giving your answer in the form $y = mx + c$

(4)

The finite region R is bounded by C, l and the y-axis.

(c) Using calculus, find the area of R.
Give your answer to one decimal place.

(7)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

P 7 1 8 1 9 A 0 3 2 3 6

5. Question 11 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 11 · part a.i · 0 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

11

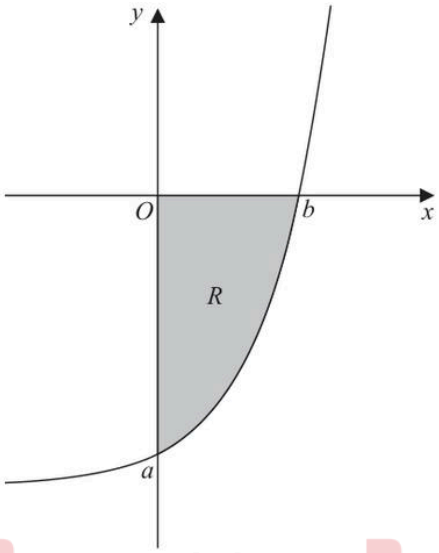


Diagram **NOT** accurately drawn

Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the curve with equation $y = e^{2x} - 9$ and the coordinate axes.

The curve crosses the coordinate axes at the points with coordinates $(0, a)$ and $(b, 0)$

(a) (i) Find the value of a

(ii) Show that $b = \ln 3$

(3)

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.

Give your answer in the form $\pi(p \ln 3 + q)$, where p and q are integers.

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6. Question 11 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 7 · part a · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

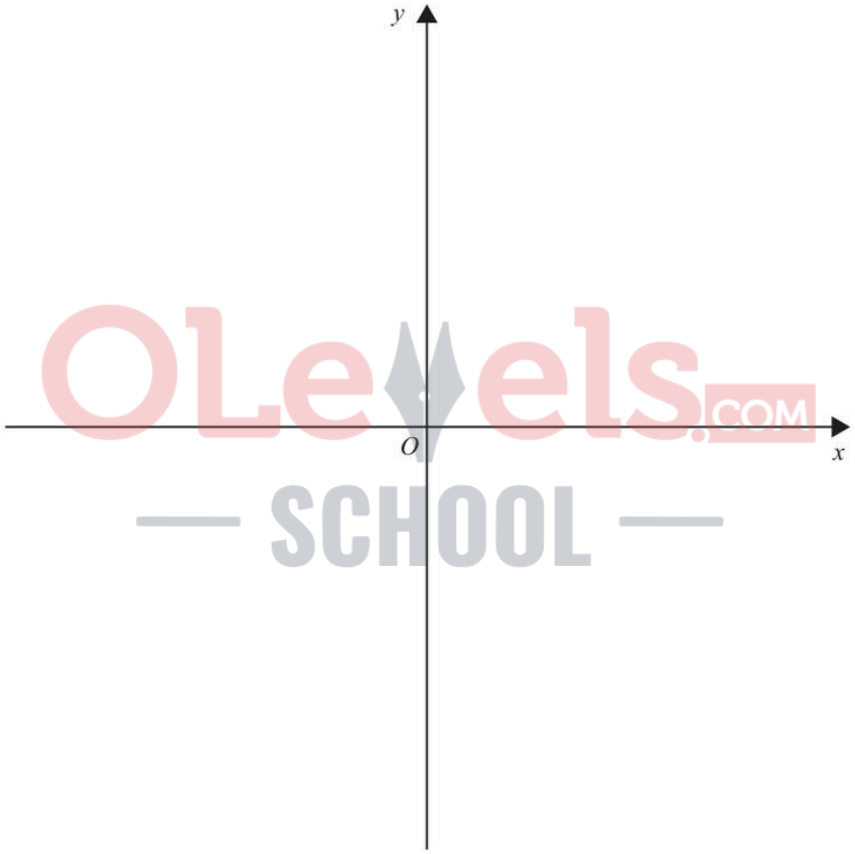
7 The curve C has equation $y = -\log_4(x + 4)$

(a) Using the axes below, sketch the graph of C.
Label the coordinates of the points of intersection of C with the coordinate axes and the equation of any asymptote to C.

(4)

(b) Solve the equation $\log_{(x+4)} 256 - \log_4(x + 4) = 0$

(5)



OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

7. Question 7 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 7 · part a · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

7 A curve C has equation $y = \log_{10}(x + 2)$

(a) Using the axes below, sketch the graph of C.
Label the coordinates of the points of intersection of C with the coordinate axes.

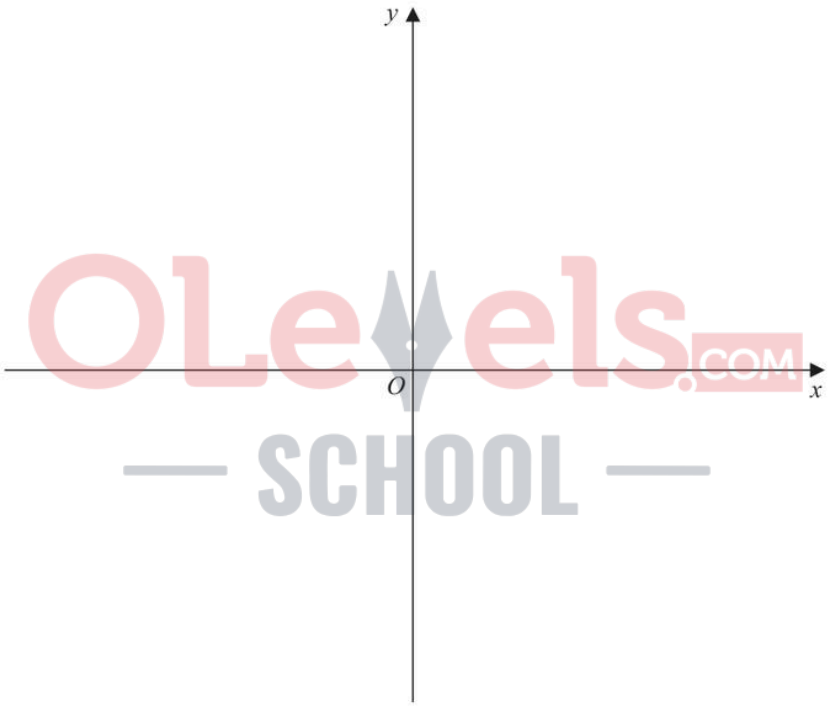
(2)

(b) Solve the equation $2(\log_a 4 + \log_a 16) = 1$

(3)

(c) Solve the equation $5\log_q 16 + 4\log_2 q = 24$

(6)



OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 6 · part a · 3 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

6

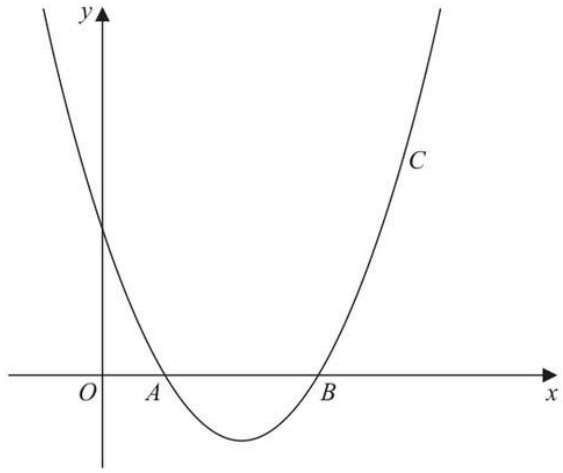


Diagram **NOT** accurately drawn

Figure 3

The curve C with equation $y = x^2 - 5x + 4$ crosses the x-axis at the points A and B, as shown in Figure 3

(a) Find the coordinates of A and the coordinates of B.

(3)

The tangent to C at A meets the tangent to C at B at the point T.

(b) Find the coordinates of T.

(6)

The normal to C at A meets the normal to C at B at the point N.

(c) Find the coordinates of N.

(3)

(d) Find the area of the quadrilateral $ATBN$.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

9. Question 6 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 3 · part c · 2 marks

2023 JAN · Paper 02

Further practice · Attempt part c only.

3

$$f(x) = 8x^2 + 10x - 3$$

Given that $f(x)$ can be written in the form $A(x + B)^2 + C$ where A, B and C are constants,

(a) find the value of A, the value of B and the value of C.

(3)

(b) Hence, or otherwise, find,

(i) the value of x for which $f(x)$ has a minimum,

(ii) the minimum value of $f(x)$.

(2)

The curve C has equation $y = f(x)$.

(c) Find the x coordinate of each of the points where C crosses the x-axis.

(2)

The straight line l has equation $y = 2x + 13$

(d) Use algebra to find the coordinates of the two points of intersection of C and l.

(4)

Using the same axes and the results of parts (b), (c) and (d),

(e) sketch the curve C and the straight line l.

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 3 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 5 · part a · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

5

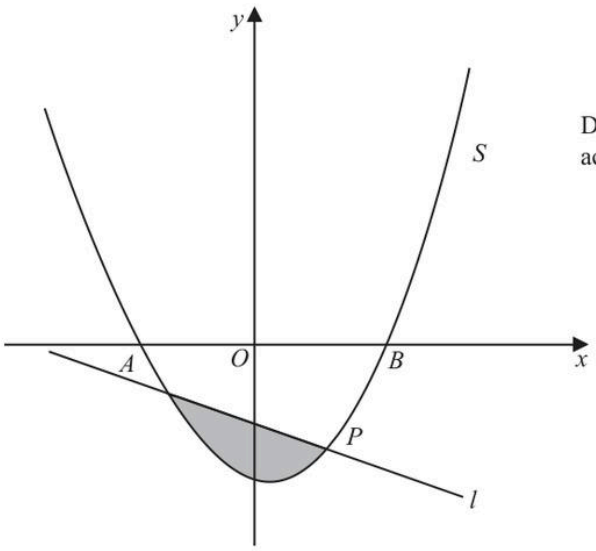


Figure 1

Figure 1 shows part of the curve S with equation $y = px^2 + qx + r$ where p, q and r are constants.

The points A, B and P with coordinates $(-2, 0)$, $(6, 0)$ and $(4, -6)$ respectively lie on S

(a) Show that an equation of S is $y = \frac{x^2}{2} - 2x - 6$

(3)

The line l is the normal to S at the point P

(b) Show that an equation of l is $2y + x + 8 = 0$

(5)

The finite region shown shaded in Figure 1 is bounded by S and l

(c) Use algebraic integration to find the exact area of the shaded region.

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

P 7 3 5 8 6 A 0 1 2 3 2

11. Question 5 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Solving an equation with your graph and a straight line

Rearranging a polynomial, reciprocal or trigonometric equation so that one side is the plotted curve, drawing the matching straight line and reading off the roots.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-GRA-08

1. Question 8 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

8 (a) Complete the table of values for $y = 2x + \frac{3}{x^2} - 3$ giving your answers to 2 decimal places where appropriate.

x	0.5	0.75	1	1.5	2	3	4	5
y	10		2				5.19	7.12

(2)

(b) On the grid opposite, draw the graph of $y = 2x + \frac{3}{x^2} - 3$ for $0.5 \leqslant x \leqslant 5$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $4x^3 - 10x^2 + 3 = 0$ for $0.5 \leqslant x \leqslant 5$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
OLevels.com SCHOOL

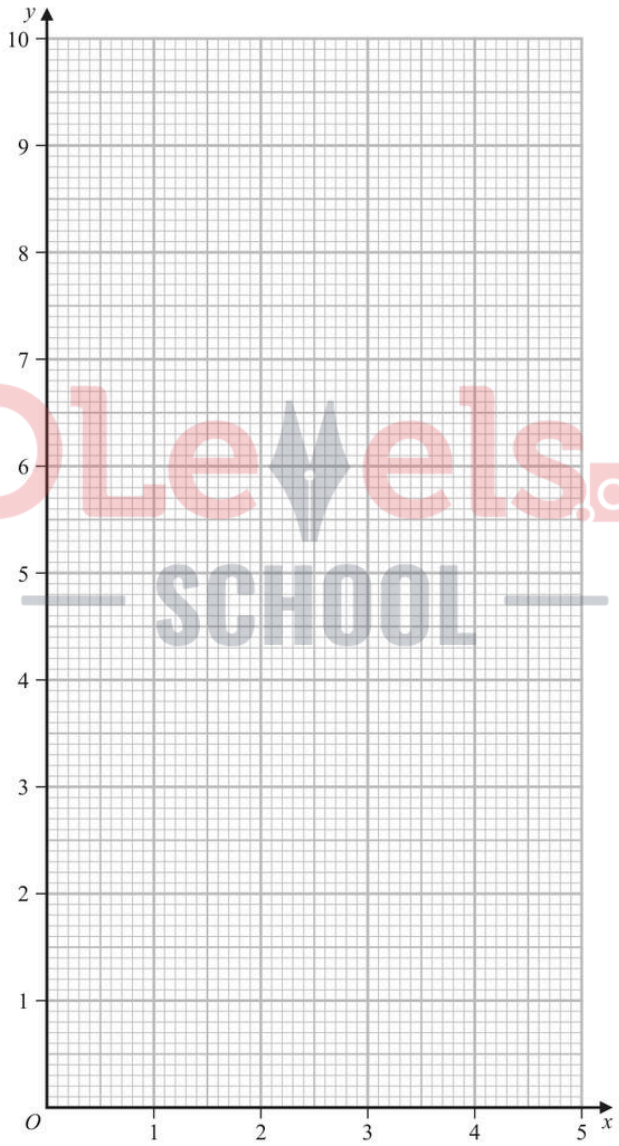
P 6 6 0 2 7 R R A 0 2 2 3 6

1. Question 8 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 8 continued



Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

P 6 6 0 2 7 R R A 0 2 3 3 6

23

Turn over ▶

1. Question 8 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

24

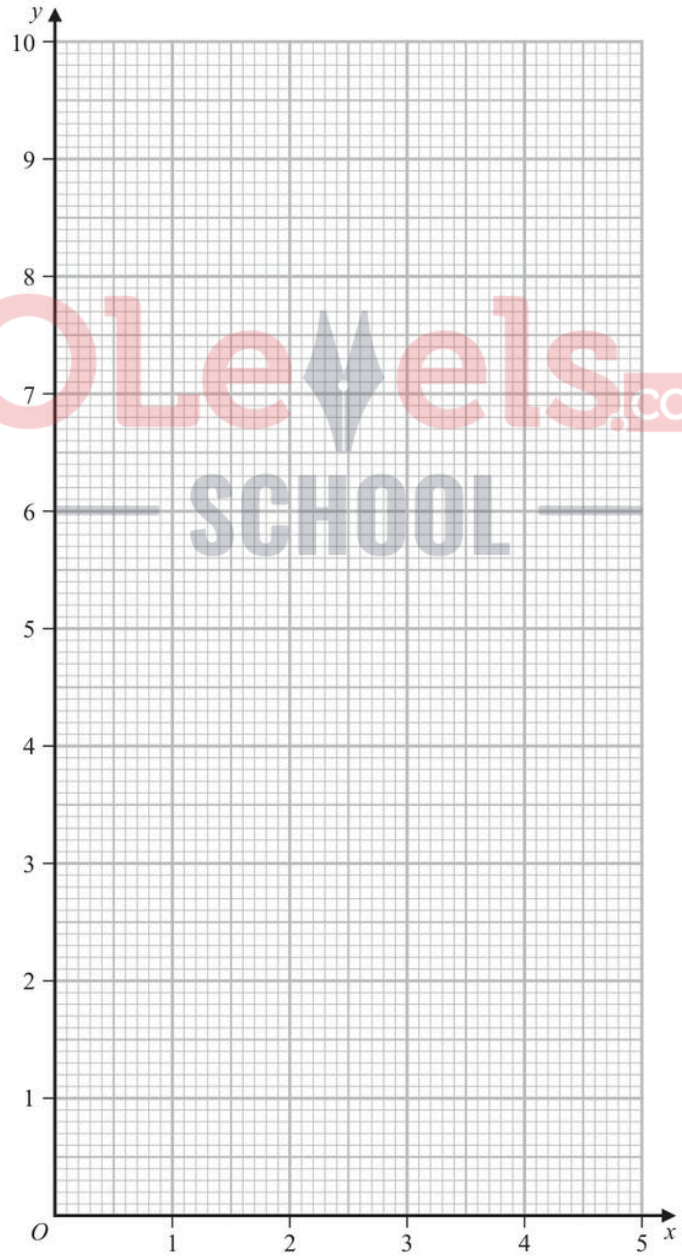
1. Question 8 · part c · 5 marks

2021 NOV · Paper 02

Worked-example retry · Attempt part c only.

Question 8 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 8 is 9 marks)

1. Question 8 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 8 · part c · 3 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

8 (a) Complete the table of values for $y = 2 + \ln(2x + 1)$ giving your answers to 2 decimal places.

(2)

x	0	0.25	0.5	1	1.5	2	3
y	2			3.10	3.39	3.61	

(b) On the grid opposite, draw the graph of $y = 2 + \ln(2x + 1)$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $\ln(2x + 1) = 3x - 4$ in the interval $0 \leqslant x \leqslant 3$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $e^{(6-x)} - (2x + 1)^2 = 0$ in the interval $0 \leqslant x \leqslant 3$

(4)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

20

P 6 1 8 8 2 A 0 2 0 3 6

2. Question 8 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

4

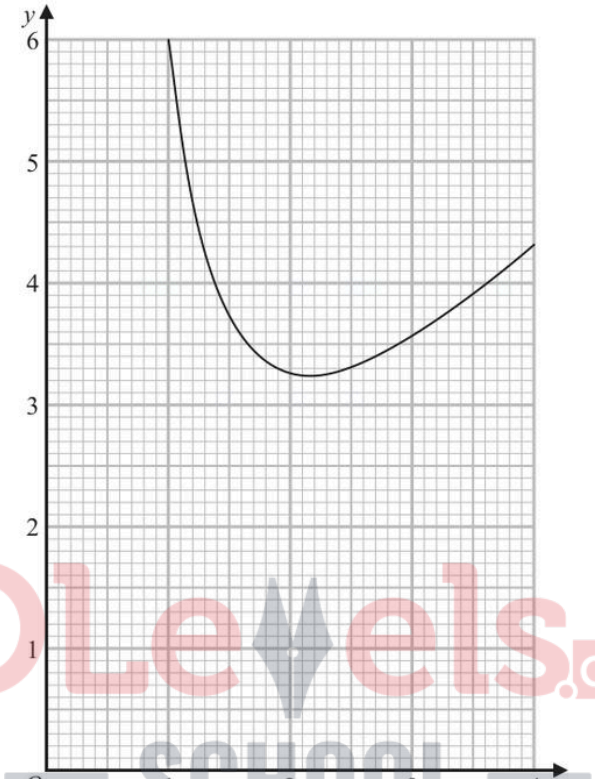


Figure 2

Figure 2 shows the graph of $y = x + \frac{5}{x^2}$ for $1 \leqslant x \leqslant 4$ drawn on a grid.

(a) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, for the roots of the equation

$$x^3 - 4x^2 + 5 = 0$$

in the interval $1 \leqslant x \leqslant 4$

(3)

(b) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, for the root of the equation

$$x^3 - x^2 - 5 = 0$$

in the interval $1 \leqslant x \leqslant 4$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

3. Question 4 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 4 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

4

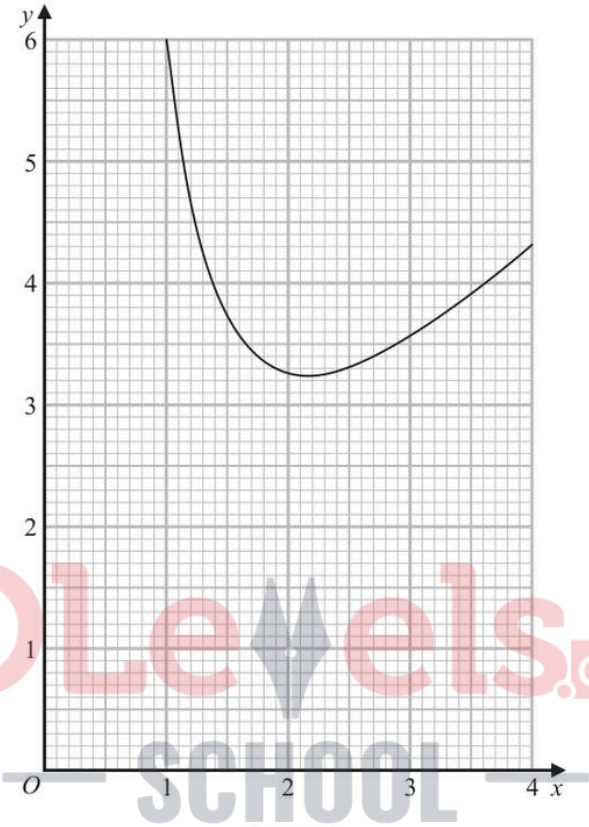


Figure 2

Figure 2 shows the graph of $y = x + \frac{5}{x^2}$ for $1 \leqslant x \leqslant 4$ drawn on a grid.

(a) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, for the roots of the equation

$$x^3 - 4x^2 + 5 = 0$$

in the interval $1 \leqslant x \leqslant 4$

(3)

(b) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, for the root of the equation

$$x^3 - x^2 - 5 = 0$$

in the interval $1 \leqslant x \leqslant 4$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

4. Question 4 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 4 is 7 marks)

4. Question 4 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part c · 2 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part c only.

4 (a) Complete the table of values for $y = 2x + 1 + \frac{2}{x^2}$

Give your answers to 2 decimal places where appropriate.

x	0.5	1	1.5	2	2.5	3	3.5
y		5			6.32		8.16

(2)

(b) On the grid opposite, draw the graph of $y = 2x + 1 + \frac{2}{x^2}$ for $0.5 \leqslant x \leqslant 3.5$

(2)

(c) Use your graph to obtain estimates, to 1 decimal place, of the roots of the equation

$$2x + \frac{2}{x^2} = 7 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(2)

(d) By drawing a suitable straight line on the grid, obtain estimates, to 1 decimal place, of the roots of the equation

$$\frac{3x}{2} + \frac{2}{x^2} = 5 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

5. Question 4 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 4 · part d · 5 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

4 (a) Complete the table of values for $y = 2x + 1 + \frac{2}{x^2}$

Give your answers to 2 decimal places where appropriate.

x	0.5	1	1.5	2	2.5	3	3.5
y		5			6.32		8.16

(2)

(b) On the grid opposite, draw the graph of $y = 2x + 1 + \frac{2}{x^2}$ for $0.5 \leqslant x \leqslant 3.5$

(2)

(c) Use your graph to obtain estimates, to 1 decimal place, of the roots of the equation

$$2x + \frac{2}{x^2} = 7 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(2)

(d) By drawing a suitable straight line on the grid, obtain estimates, to 1 decimal place, of the roots of the equation

$$\frac{3x}{2} + \frac{2}{x^2} = 5 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

6. Question 4 · part d · 5 marks

2020 NOV · Paper 02 (regional R)

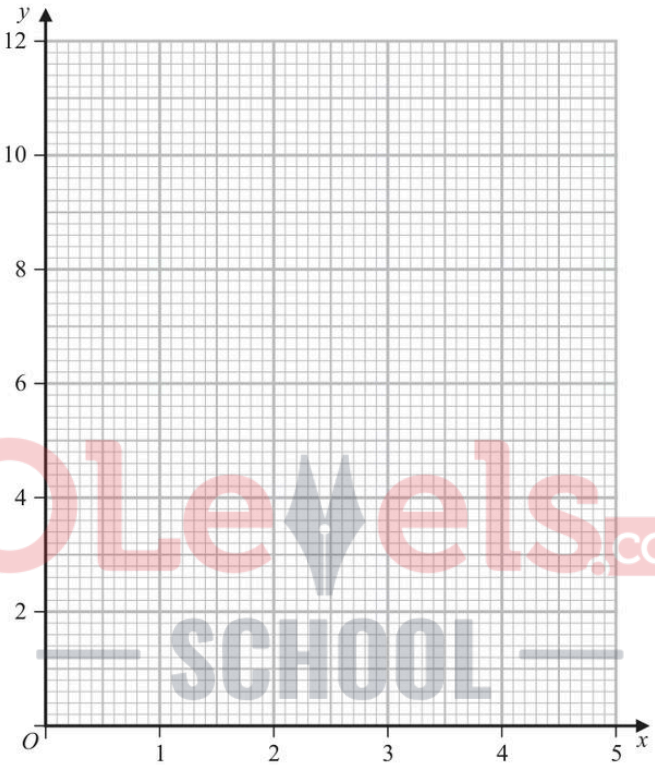
Further practice · Attempt part d only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued



OLevels.COM SCHOOL

Turn over for a spare grid if you need to redraw your graph.

P 6 2 2 8 4 A 0 9 3 2

9

Turn over ▶

6. Question 4 · part d · 5 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10

P 6 2 2 8 4 A 0 1 0 3 2

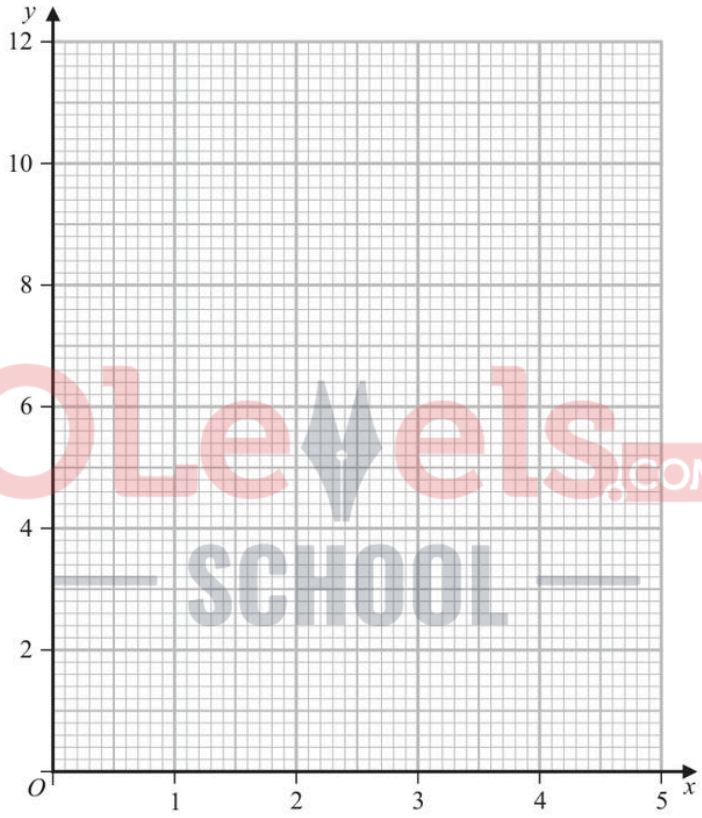
6. Question 4 · part d · 5 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 4 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 4 is 11 marks)

6. Question 4 · part d · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part d · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.

8 (a) Given that x is measured in radians, complete the table of values for

$$y = \sin x - 3\cos 2x - 0.5$$

Give your answers to one decimal place.

x	0	0.5	0.8	1	1.6	2	2.5	3
y			0.3	1.6			–0.8	–3.2

(2)

(b) On the grid opposite, draw the graph of $y = \sin x - 3\cos 2x - 0.5$ for $0 \leqslant x \leqslant 3$

(2)

(c) Use a formula from page 2 to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

$$f(x) = 2\sin x + 12\sin^2 x - x - 5$$

(d) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $f(x) = 0$ in the interval $0 \leqslant x \leqslant 3$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

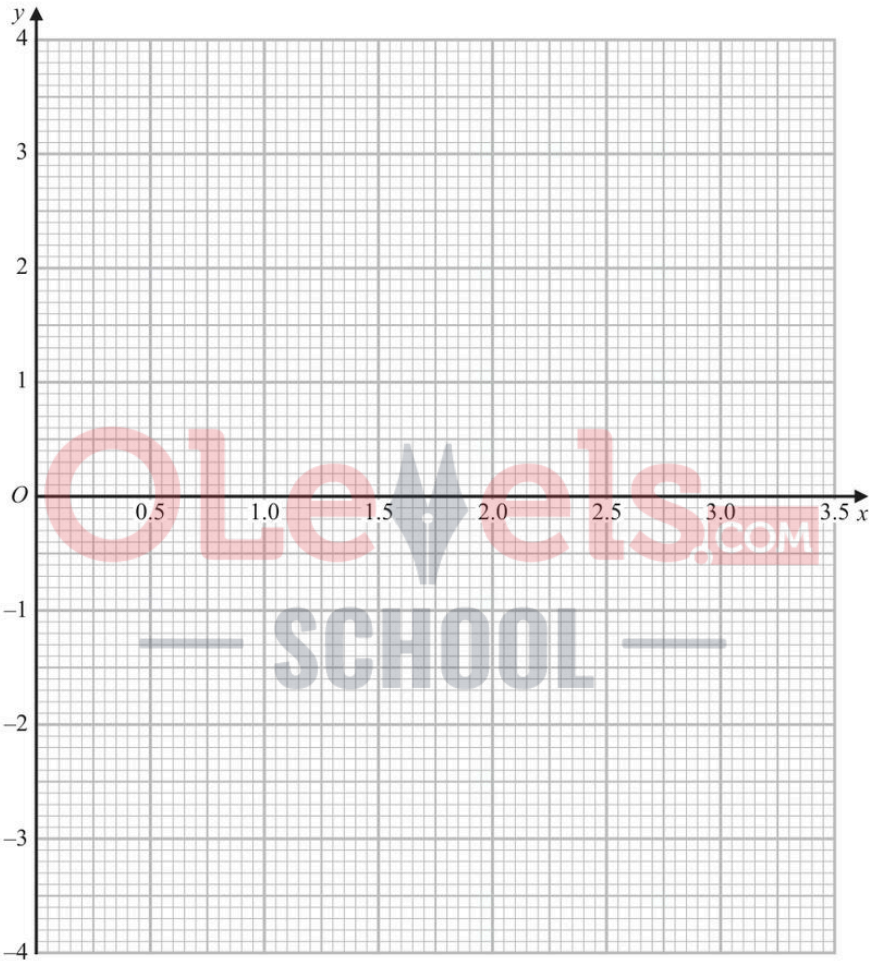
P 6 6 3 0 8 A 0 2 4 3 6

7. Question 8 · part d · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.

Question 8 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Turn over for a spare grid if you need to redraw your graph.

7. Question 8 · part d · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.



Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

26

P 6 6 3 0 8 A 0 2 6 3 6

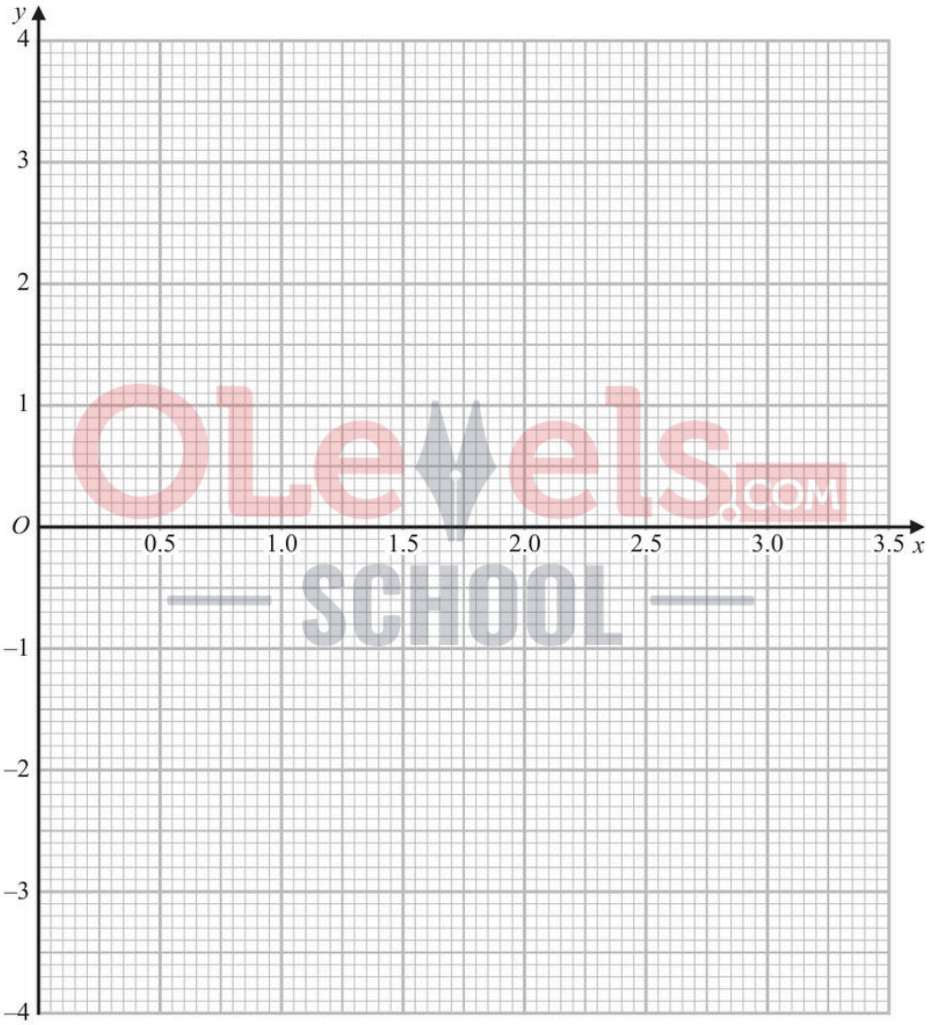
7. Question 8 · part d · 5 marks

2022 JAN · Paper 02

Further practice · Attempt part d only.

Question 8 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 8 is 11 marks)

7. Question 8 · part d · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 6 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 (a) Complete the table of values for

$$y = x - \frac{3}{x^2}$$

giving your answers to one decimal place where appropriate.

x	0.5	1	1.5	2	3	4	5	6
y	–11.5			1.3	2.7			5.9

(2)

(b) On the grid opposite, draw the graph of $y = x - \frac{3}{x^2}$ for $0.5 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of each of the two roots of the equation

$$2x^3 - 6x^2 + 3 = 0$$

in the interval $0.5 \leqslant x \leqslant 6$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

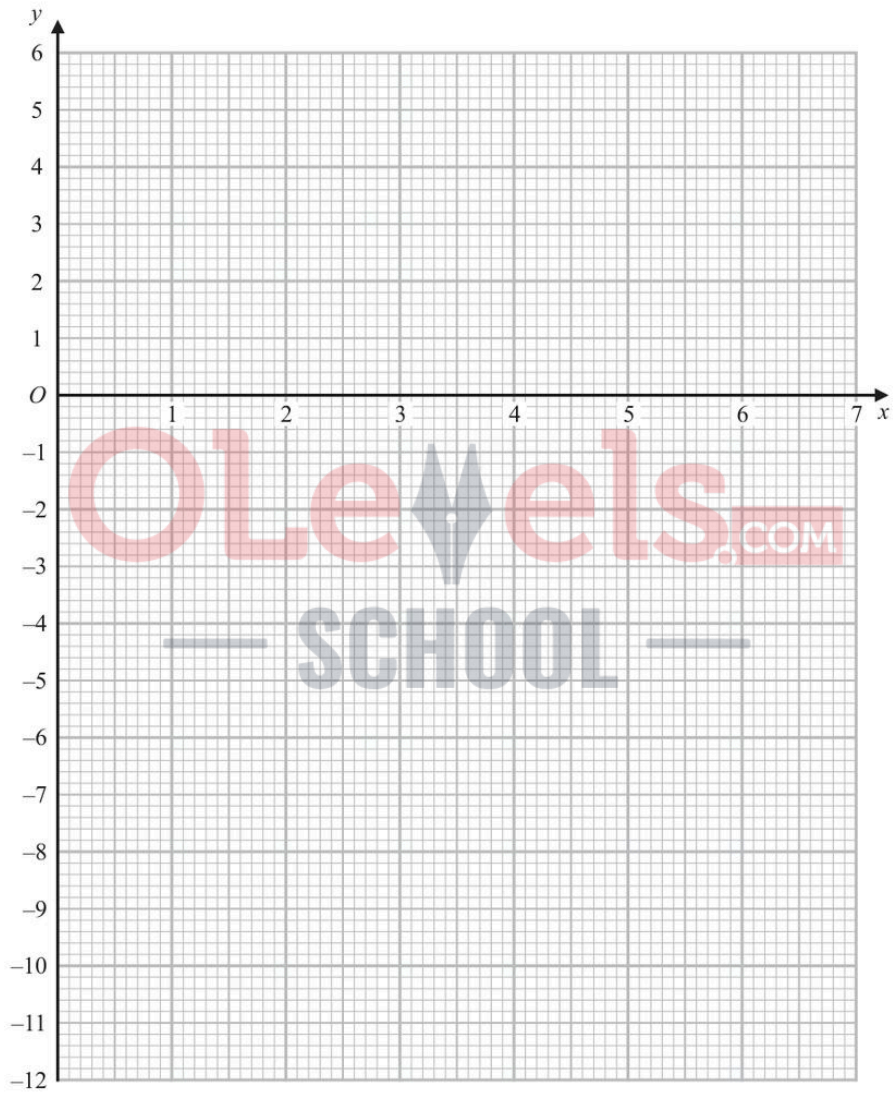
P 6 0 4 7 5 A 0 1 2 3 6

8. Question 6 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued



OLevels.com SCHOOL

Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

8. Question 6 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

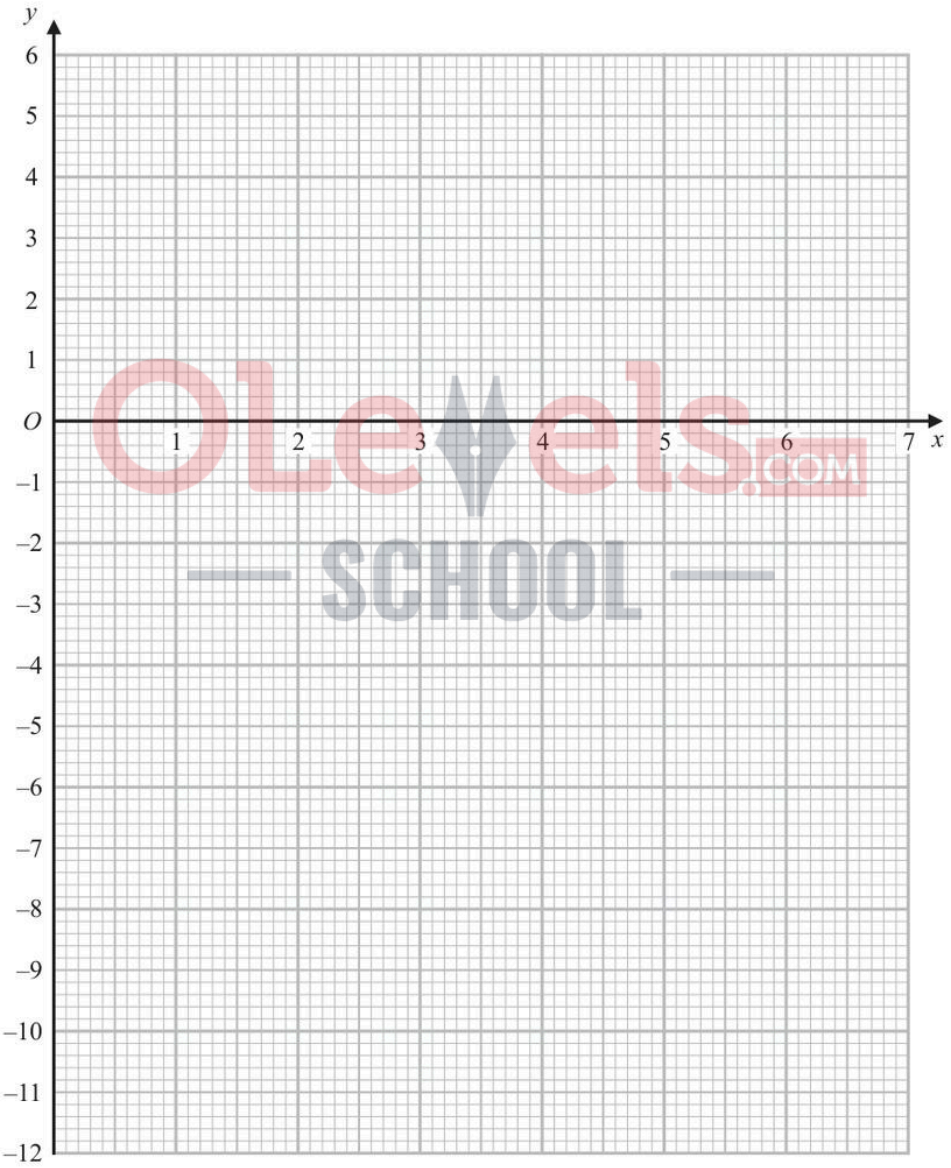
8. Question 6 · part c · 5 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 6 is 9 marks)

8. Question 6 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 6 · part c · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part c only.

6 (a) Complete the table of values for $y = 1 + 3e^{-x}$ giving your answers to 2 decimal places where appropriate.

(2)

x	0	0.25	0.5	1	1.5	2	3
y		3.34	2.82		1.67		1.15

(b) On the grid opposite, draw the graph of $y = 1 + 3e^{-x}$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$x = e^{-x} \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\ln(x-1)^3 = -3x \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(4)

OLevels.COM — SCHOOL —

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

9. Question 6 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 3 · part whole · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

3

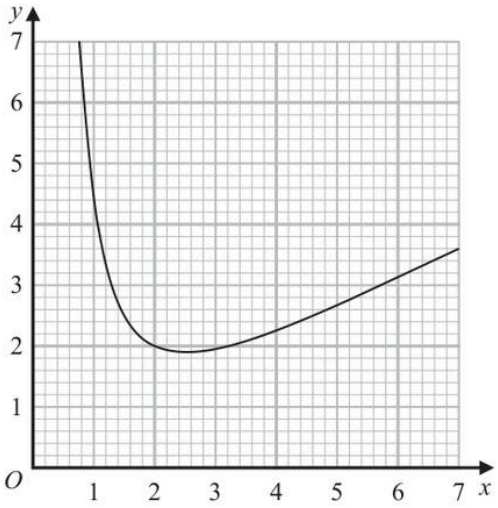


Figure 1

Figure 1 shows part of the curve with equation $y = \frac{x}{2} + \frac{4}{x^2}$ in the interval $0.8 < x < 7$

By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the roots of the equation $3x^3 - 12x^2 + 8 = 0$ in the interval $0.8 < x < 7$

(5)

OLevels.com — SCHOOL —

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

10. Question 3 · part whole · 5 marks

2023 JUN · Paper 01

Further practice · Attempt part whole only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 3 is 5 marks)

10. Question 3 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 4 · part whole · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

4

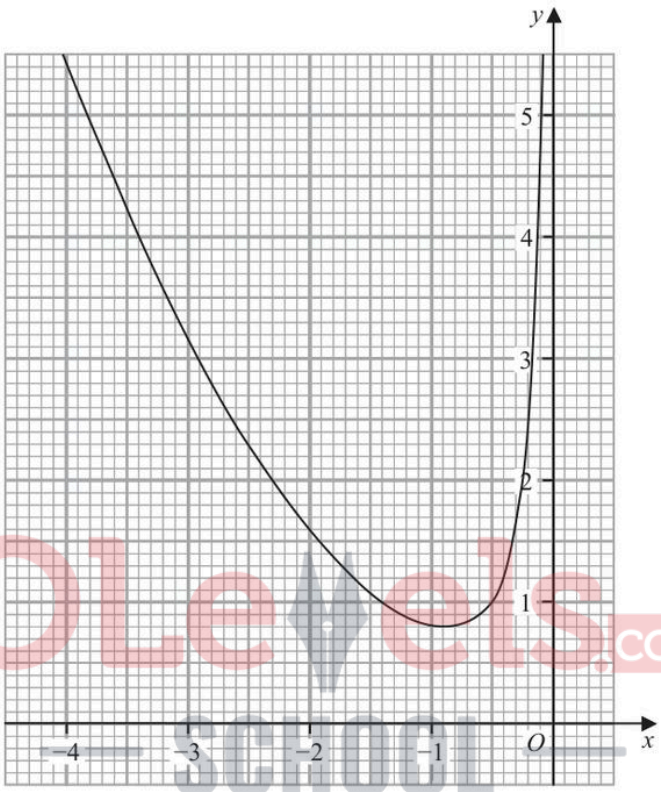


Figure 2

Figure 2 shows part of the curve with equation $y = \frac{x^2}{3} - \frac{1}{2x}$ for $-4 < x < 0$

By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $4x^3 + 3x^2 - 36x - 6 = 0$ in the interval $-4 < x < 0$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

8

P 7 6 5 0 8 A 0 8 3 2

11. Question 4 · part whole · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 4 continued

OLevels.com SCHOOL

(Total for Question 4 is 4 marks)

11. Question 4 · part whole · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 9 · part c · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

9 (a) Complete the table of values for

$$y = \frac{x^3 + 4}{5 - x}$$

giving your answers to 2 decimal places.

x	−2	−1.5	−1	−0.5	0	0.5	1	1.5	2
y	−0.57	0.10	0.5		0.8		1.25		4

(2)

(b) On the grid opposite, draw the graph of

$$y = \frac{x^3 + 4}{5 - x} \quad \text{for } -2 \leqslant x \leqslant 2$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $x^3 - x^2 + 8x - 11 = 0$ in the interval $-2 \leqslant x \leqslant 2$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

24

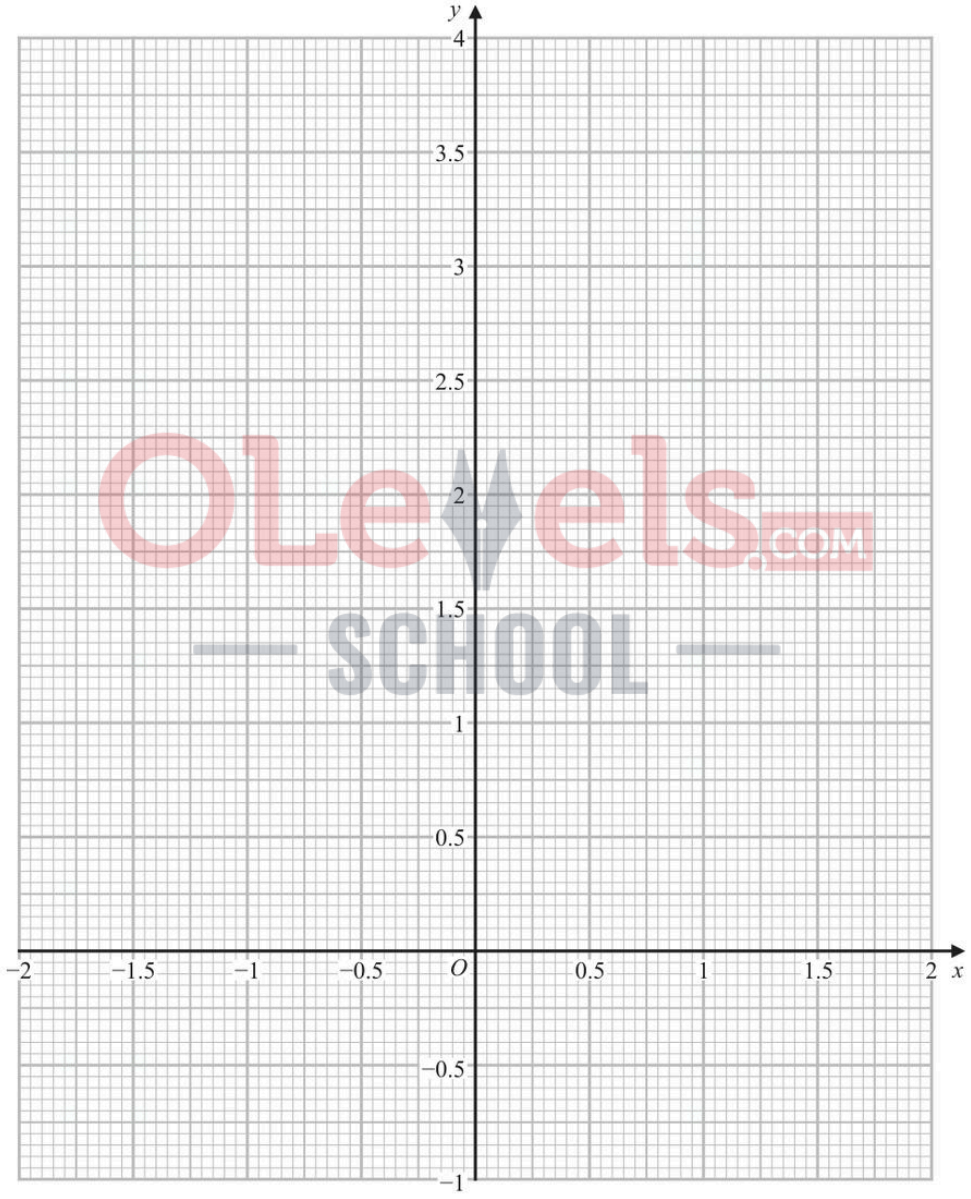
P 6 6 3 0 9 A 0 2 4 3 6

12. Question 9 · part c · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 9 continued



Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

12. Question 9 · part c · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 9 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

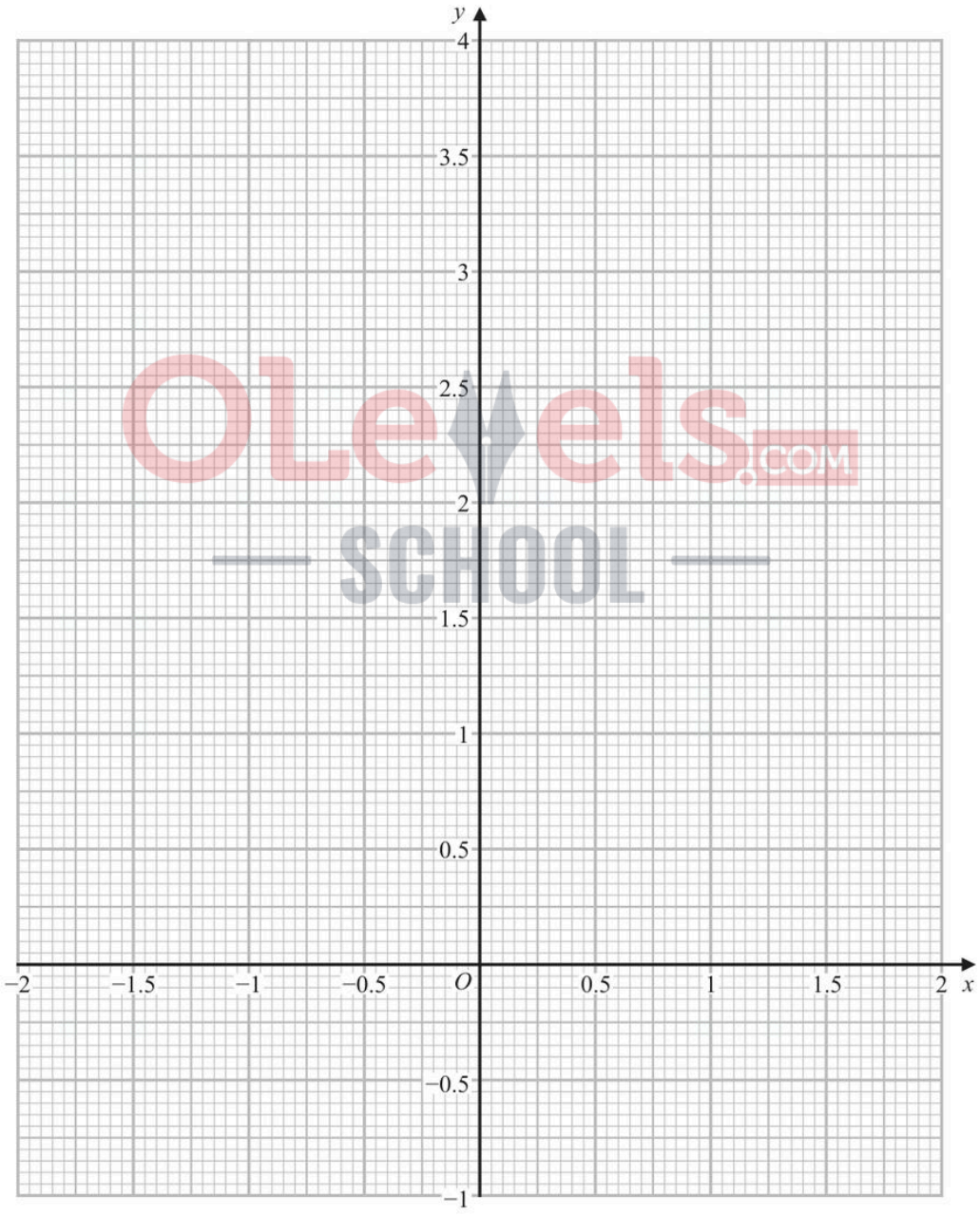
12. Question 9 · part c · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 9 continued

Only use this grid if you need to redraw your graph.



OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 9 is 9 marks)

12. Question 9 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Solving log and exponential equations with your graph

Using log laws, or switching between log and exponential form, to turn an equation into 'plotted curve = straight line', then solving it from the graph.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-GRA-09

1. Question 5 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

5 (a) Complete the table of values for $y = e^{3x-2}$ giving your answers to 2 decimal places.

x	0	0.25	0.5	0.75	1
y	0.14				2.72

(2)

(b) On the grid opposite, draw the graph of $y = e^{3x-2}$ for $0 \leqslant x \leqslant 1$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $3x = 2 + \ln(3 - x)$

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

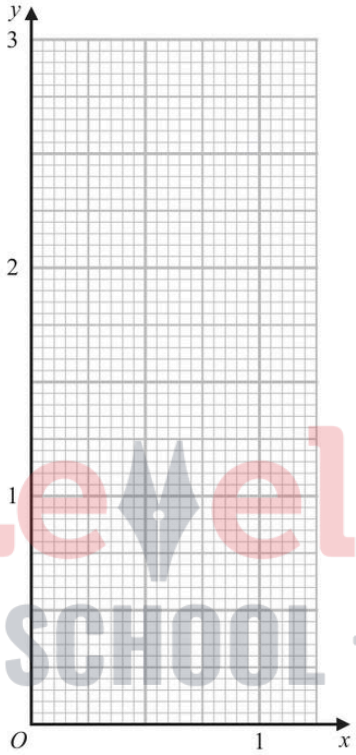
P 7 1 6 4 2 A 0 1 2 3 6

1. Question 5 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 5 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Turn over for a spare grid if you need to redraw your graph.

1. Question 5 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 5 continued

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

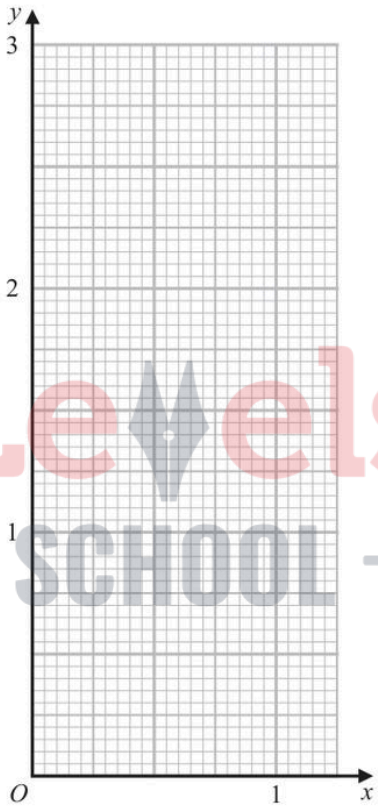
1. Question 5 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 5 continued

Only use this grid if you need to redraw your graph.



OLevels.COM SCHOOL

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 5 is 7 marks)

1. Question 5 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 8 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

8 (a) Complete the table of values for $y = 2 + \ln(2x + 1)$ giving your answers to 2 decimal places.

(2)

x	0	0.25	0.5	1	1.5	2	3
y	2			3.10	3.39	3.61	

(b) On the grid opposite, draw the graph of $y = 2 + \ln(2x + 1)$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $\ln(2x + 1) = 3x - 4$ in the interval $0 \leqslant x \leqslant 3$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $e^{(6-x)} - (2x + 1)^2 = 0$ in the interval $0 \leqslant x \leqslant 3$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

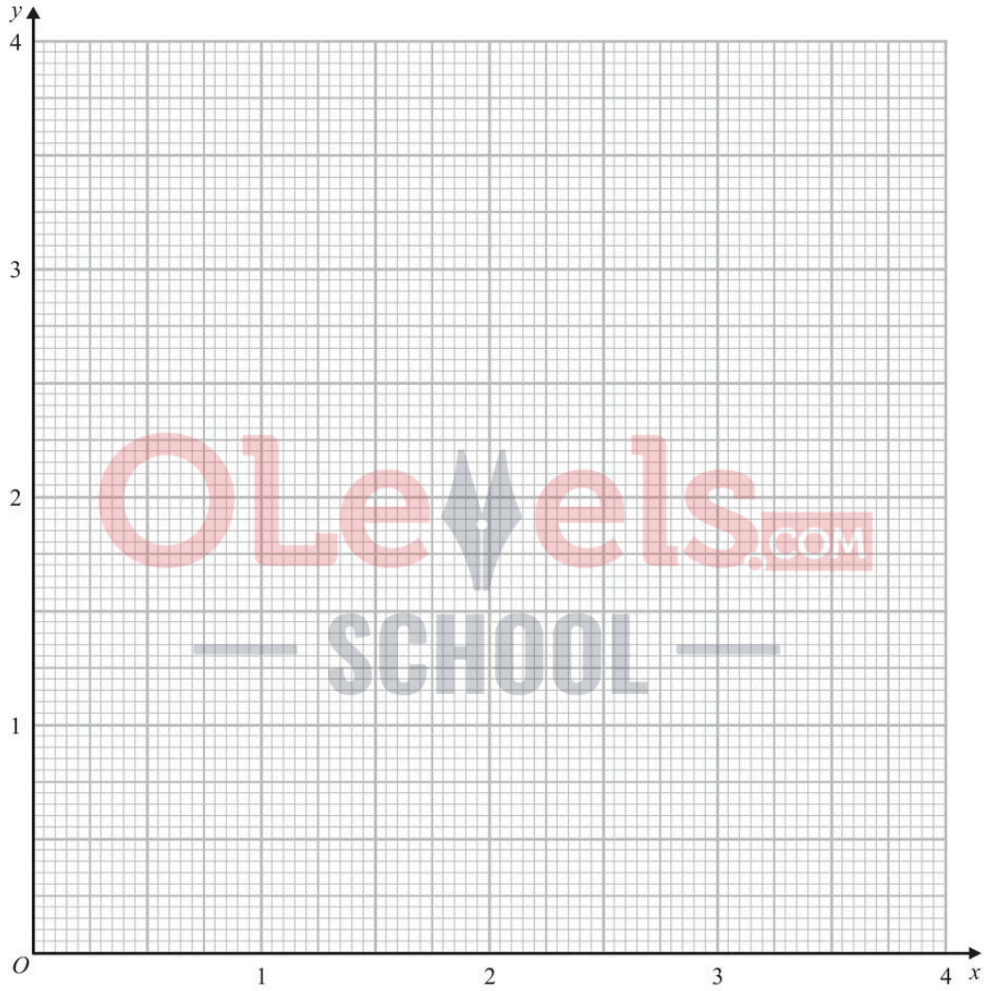
OLevels.com SCHOOL

2. Question 8 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 8 continued



Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 6 1 8 8 2 A 0 2 1 3 6

2. Question 8 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

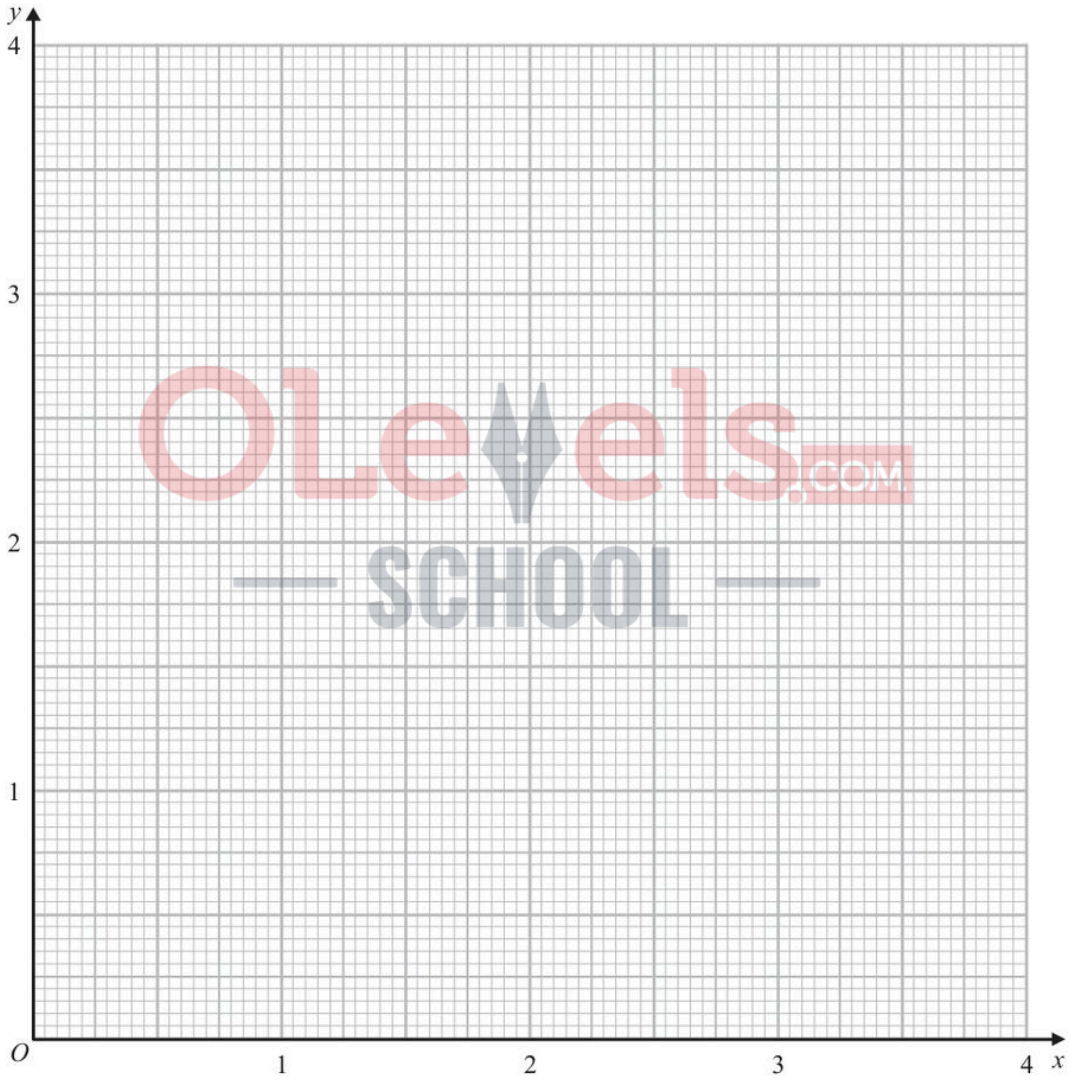
2. Question 8 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 8 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 8 is 11 marks)

2. Question 8 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part c · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

7 (a) Complete the table of values for

$$y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$$

giving each value to 2 decimal places where appropriate.

x	–6	–5	–4	–3	–2	–1	0
y	4	3.59	3.26				2.5

(2)

(b) On the grid opposite, draw the graph of $y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$ for $-6 \leqslant x \leqslant 0$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_2(2x+2)^3 + x + 3 = 0 \text{ in the interval } -6 \leqslant x \leqslant 0$$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

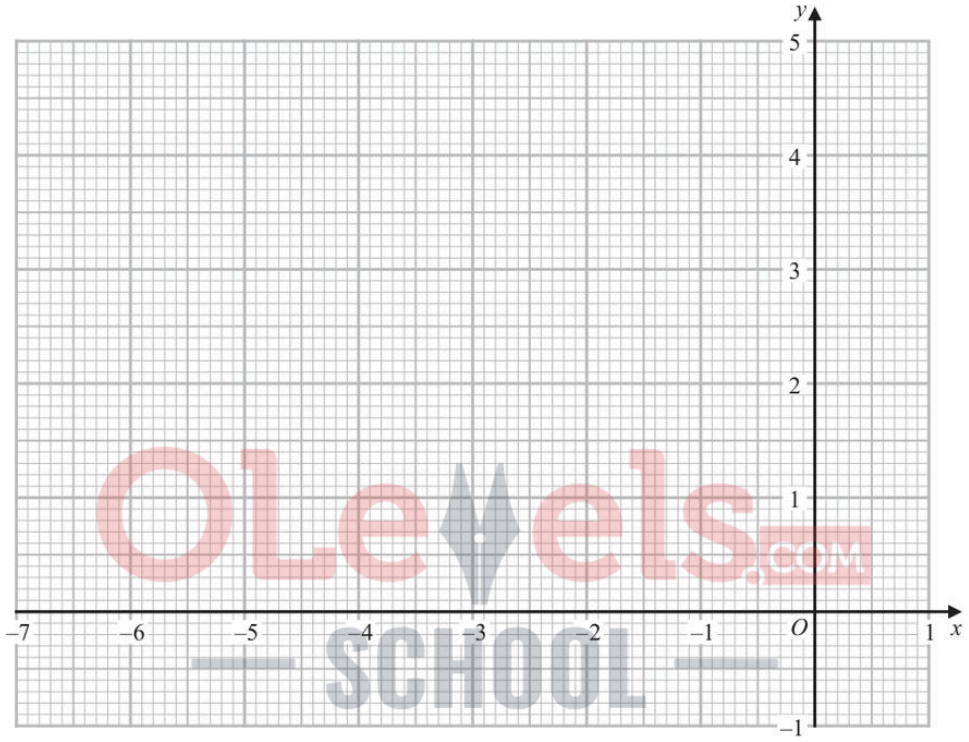
OLevels.COM SCHOOL

3. Question 7 · part c · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 7 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Turn over for a spare grid if you need to redraw your graph.

3. Question 7 · part c · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

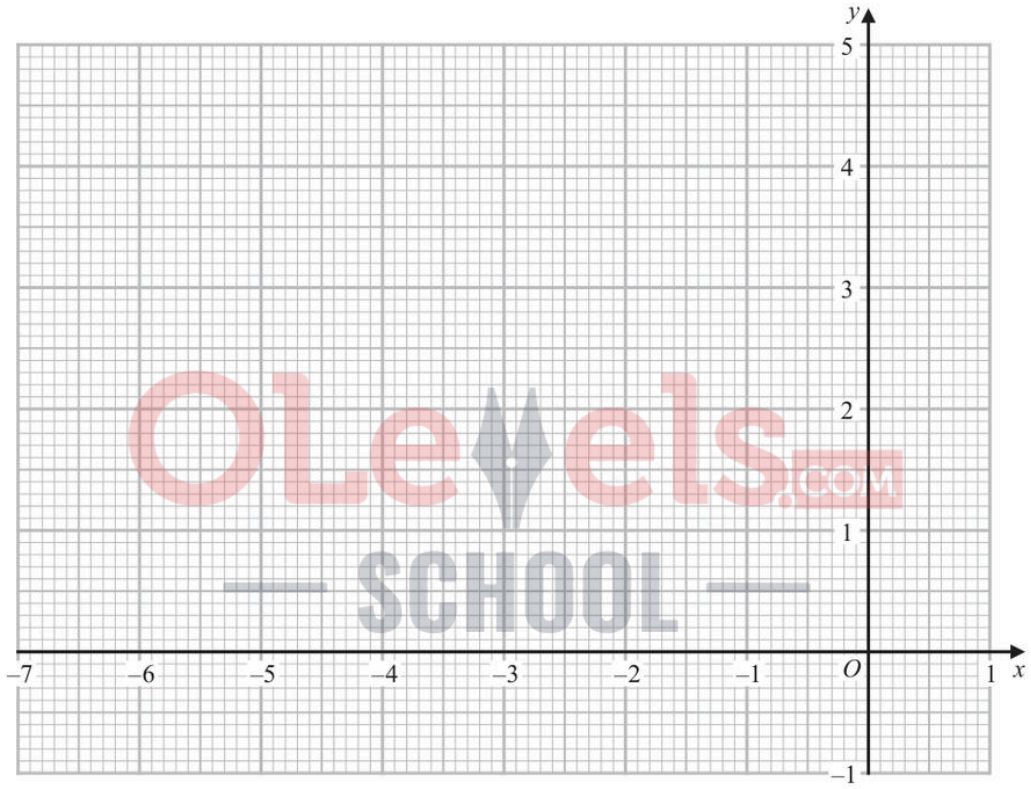
3. Question 7 · part c · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 7 continued

Only use this grid if you need to redraw your graph.



OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 7 is 10 marks)

3. Question 7 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 7 · part c · 6 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

7 (a) Complete the table of values for $y = \log_3(4-x)+3$ giving your answers to 2 decimal places.

x	0	0.5	1	1.5	2	2.5	3	3.5
y	4.26		4		3.63		3	2.37

(2)

(b) On the grid opposite, draw the graph of $y = \log_3(4-x)+3$ in the interval $0 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate,

to one decimal place, of the root of the equation $3^{2x-5} - (4-x)^3 = 0$ in the interval $0 \leqslant x \leqslant 3.5$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

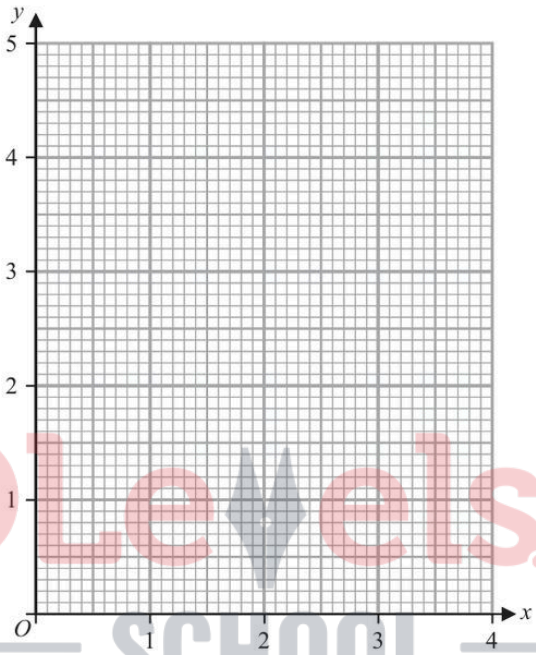
OLevels.com SCHOOL

4. Question 7 · part c · 6 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Turn over for a spare grid if you need to redraw your graph.

P 7 2 8 6 7 A 0 1 5 3 2

15

Turn over

4. Question 7 · part c · 6 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

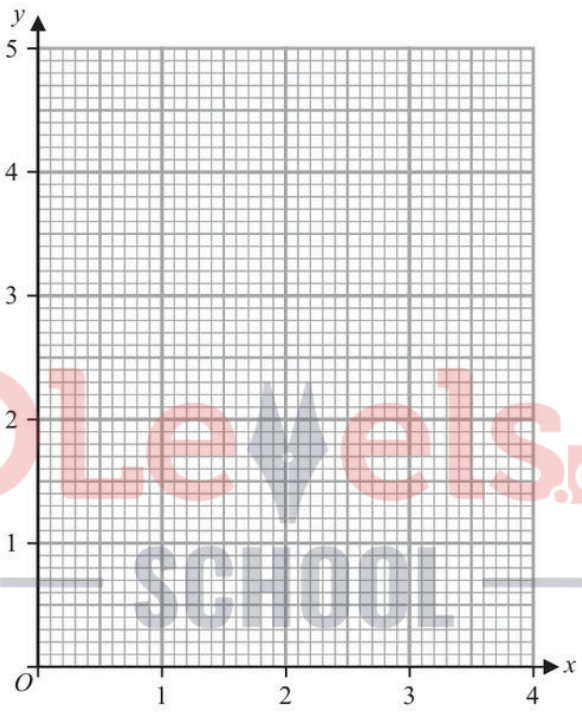
4. Question 7 · part c · 6 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

Only use this grid if you need to redraw your graph.



OLevels.COM SCHOOL

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 7 is 10 marks)

4. Question 7 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part c · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

4 (a) Complete the table of values for $y = \frac{x}{2} + 6e^{-2x} + 1$

giving your answers to one decimal place.

x	0	1	1.5	2	3	4	5	6
y	7		2.0			3.0		4.0

(2)

(b) On the grid opposite, draw the graph of $y = \frac{x}{2} + 6e^{-2x} + 1$ for $0 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on your graph, obtain estimates, to one decimal place, of the roots of the equation

$$2x + \ln(24 - 5x) = \ln 36$$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

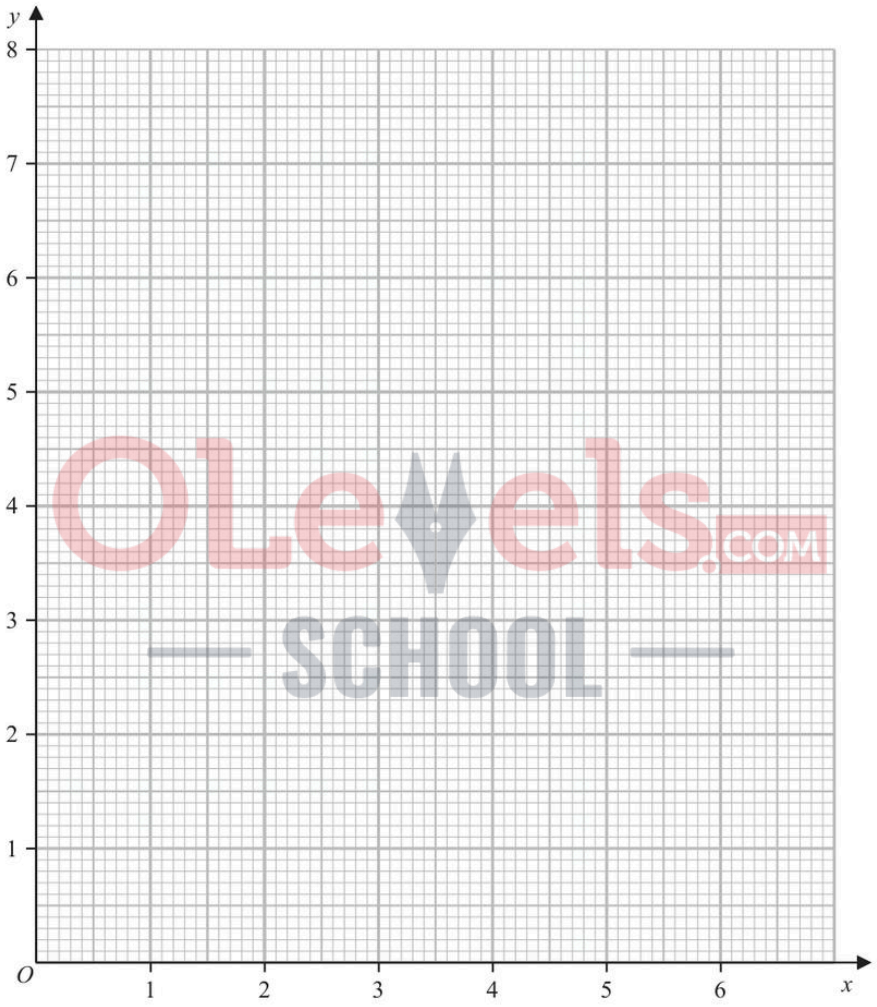
OLevels.com SCHOOL

5. Question 4 · part c · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued



OLevels.com SCHOOL

Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 4 · part c · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

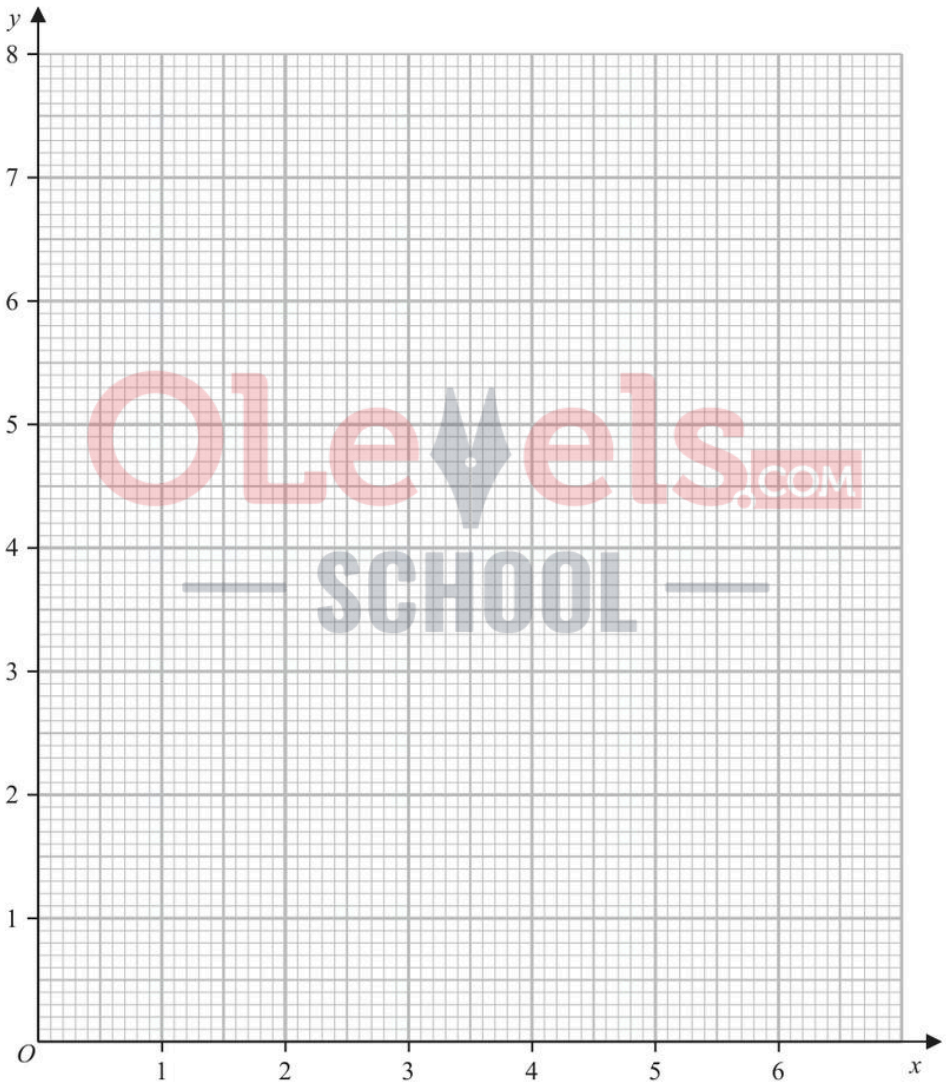
5. Question 4 · part c · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued

Only use this grid if you need to redraw your graph.



OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 4 is 9 marks)

5. Question 4 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 6 · part whole · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

6

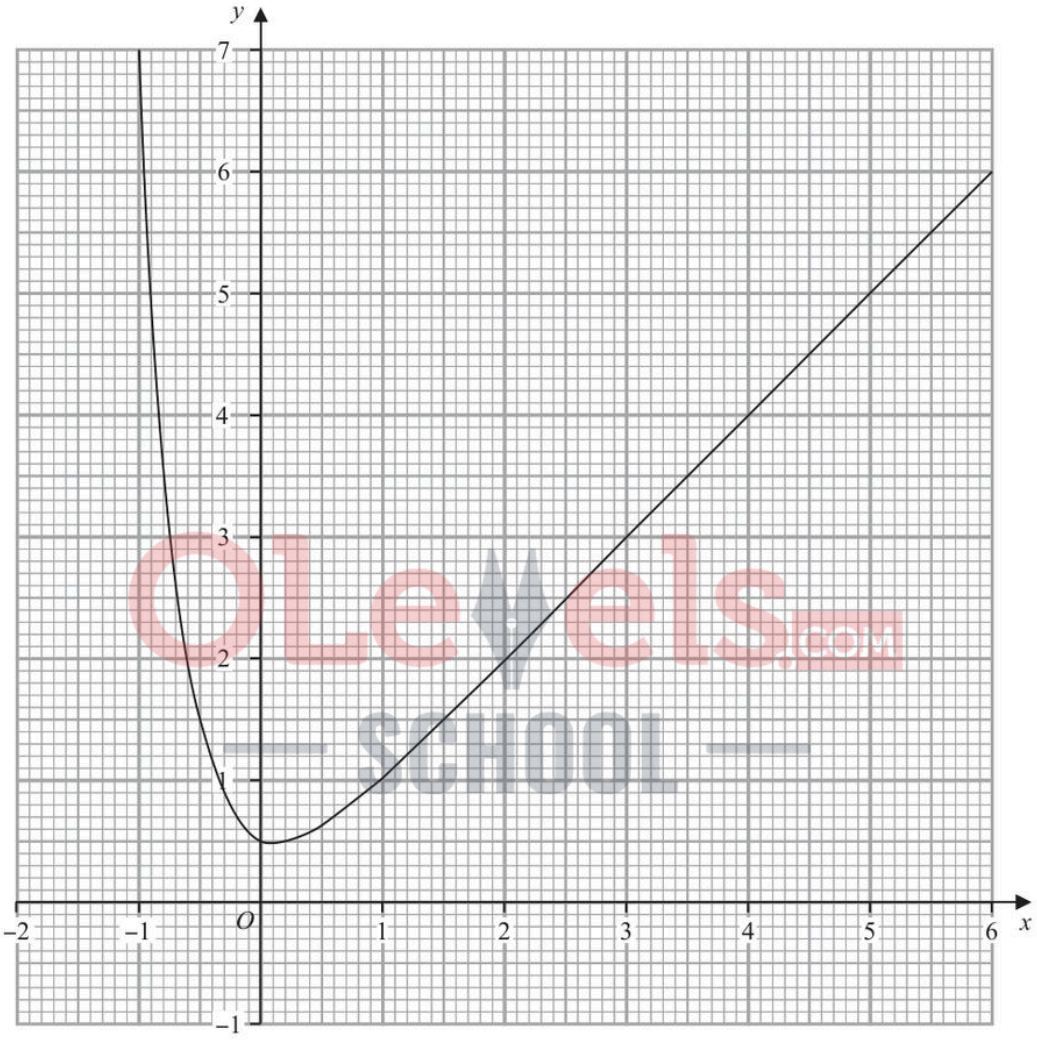


Figure 1

Figure 1 shows part of the graph of the curve with equation $y = x + 2^{-(4x+1)}$

By drawing a suitable straight line on the graph, obtain an estimate, to one decimal place, of the roots of the equation $\log_2(8 - 3x) + 4x = 0$ in the interval $-2 \leqslant x \leqslant 6$

(7)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 6 · part whole · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 6 continued

OLevels.com SCHOOL

(Total for Question 6 is 7 marks)

6. Question 6 · part whole · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part d · 5 marks

2025 NOV · Paper 01

Further practice · Attempt part d only.

8 (a) Complete the table of values for $y = 3^x - 2$ giving your answers to one decimal place.

x	-1	-0.5	0	0.25	0.5	0.75	1	1.5	2
y	-1.7		-1				1	3.2	7

(2)

(b) On the grid opposite, draw the graph of $y = 3^x - 2$ for $-1 \leqslant x \leqslant 2$

(2)

(c) Use your graph to obtain an estimate, to one decimal place, of the value of $\log_3 4.5$

Show clearly how you have used your graph.

(3)

(d) By drawing a straight line on your grid obtain an estimate, to one decimal place, of the root of the equation $\log_3(8-3x) - x = 0$ in the interval $-1 \leqslant x \leqslant 2$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

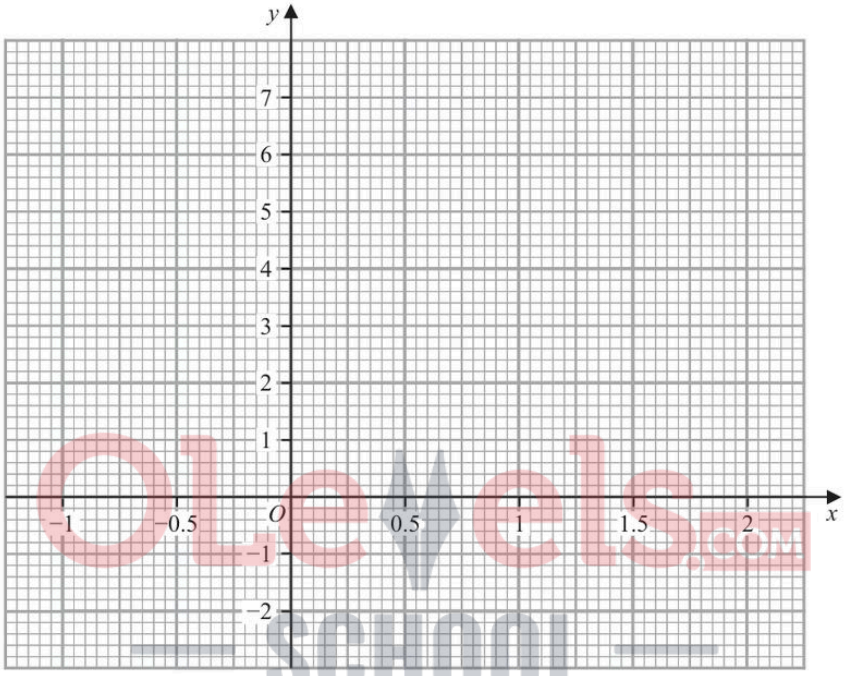
P 7 8 9 6 4 A 0 2 2 3 6

7. Question 8 · part d · 5 marks

2025 NOV · Paper 01

Further practice · Attempt part d only.

Question 8 continued



Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

7. Question 8 · part d · 5 marks

2025 NOV · Paper 01

Further practice · Attempt part d only.

Question 8 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

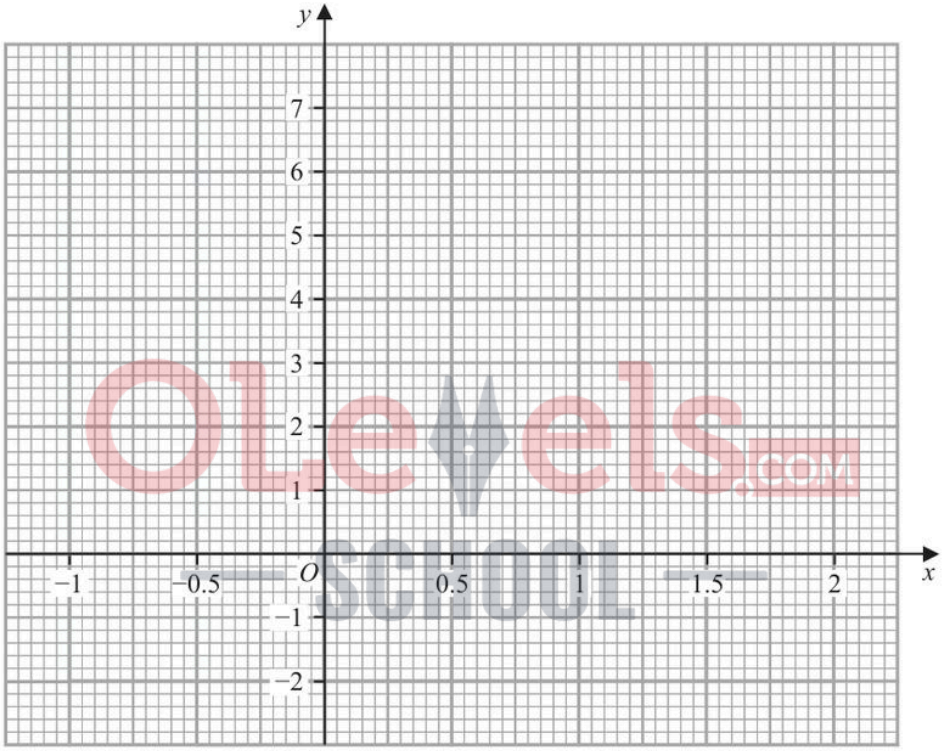
7. Question 8 · part d · 5 marks

2025 NOV · Paper 01

Further practice · Attempt part d only.

Question 8 continued

Only use this grid if you need to redraw your graph.



OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 8 is 12 marks)

7. Question 8 · part d · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 8 · part c · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part c only.

8 (a) Complete the table of values for $y = 3^x - 2$ giving your answers to one decimal place.

x	-1	-0.5	0	0.25	0.5	0.75	1	1.5	2
y	-1.7		-1				1	3.2	7

(2)

(b) On the grid opposite, draw the graph of $y = 3^x - 2$ for $-1 \leqslant x \leqslant 2$

(2)

(c) Use your graph to obtain an estimate, to one decimal place, of the value of $\log_3 4.5$

Show clearly how you have used your graph.

(3)

(d) By drawing a straight line on your grid obtain an estimate, to one decimal place, of the root of the equation $\log_3(8-3x) - x = 0$ in the interval $-1 \leqslant x \leqslant 2$

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 8 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 7 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

7 (a) Complete the table of values for $y = 3^{\frac{x}{4}} + 2$

Give your answers to 2 decimal places where appropriate.

(2)

x	0	1	2	3	4	5
y	3	3.32				5.95

(b) On the grid opposite, draw the graph of

$$y = 3^{\frac{x}{4}} + 2 \qquad \text{for } 0 \leqslant x \leqslant 5$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_3(6 - 2x)^4 - x = 0$$

in the interval $0 \leqslant x \leqslant 5$

(5)

OLevels.COM SCHOOL

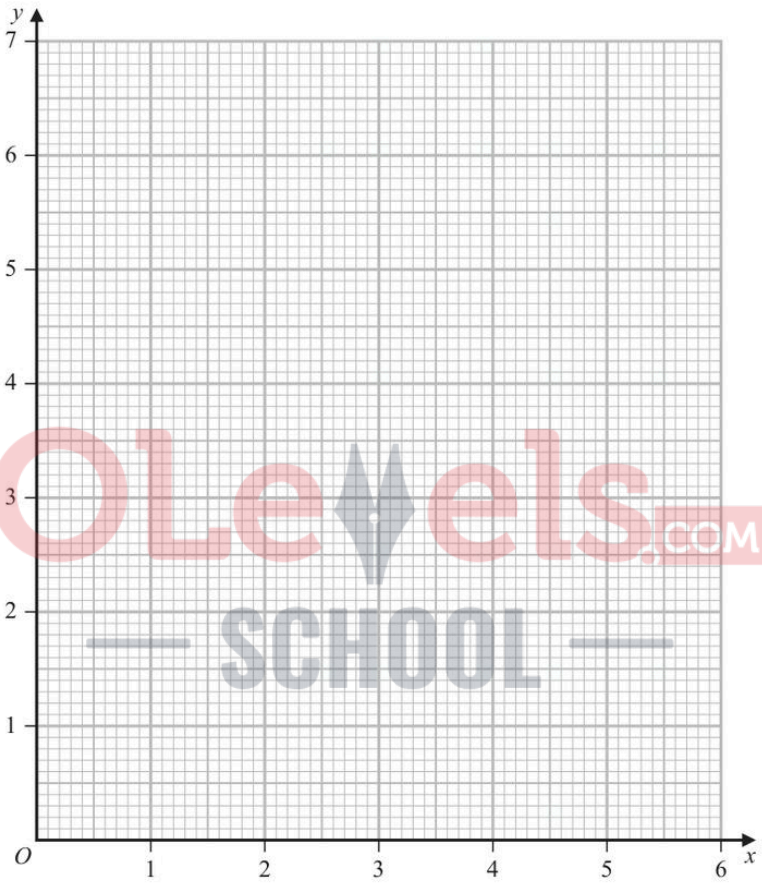
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

9. Question 7 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Turn over for a spare grid if you need to redraw your graph.

9. Question 7 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

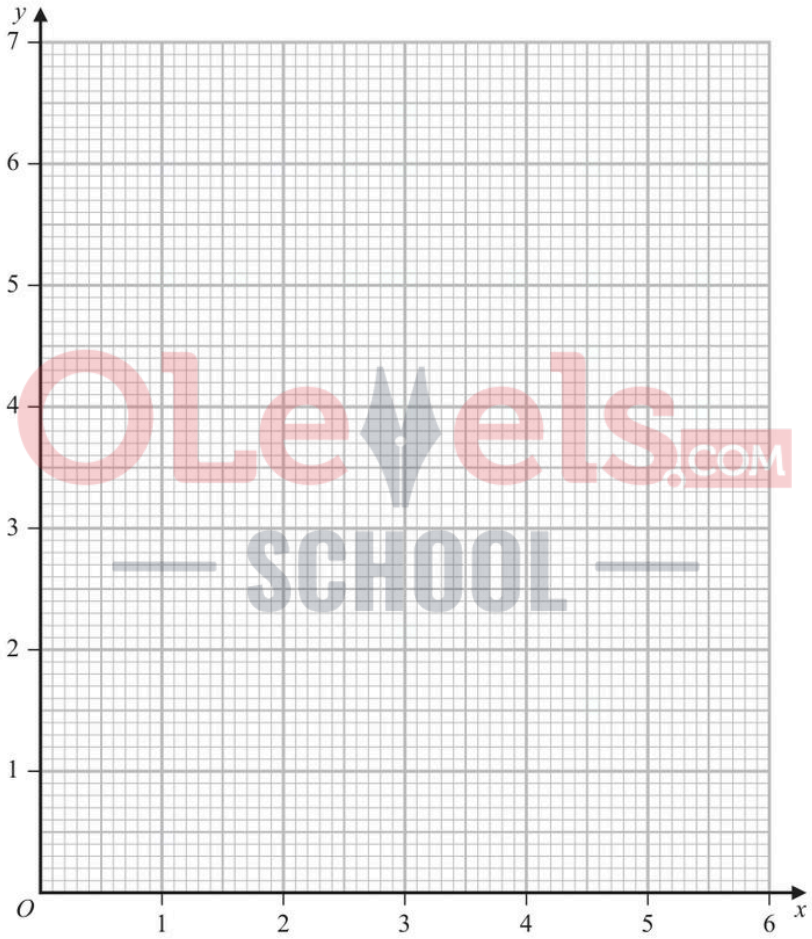
9. Question 7 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 7 is 9 marks)

9. Question 7 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 6 · part d · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

6 (a) Complete the table of values for $y = 1 + 3e^{-x}$ giving your answers to 2 decimal places where appropriate.

(2)

x	0	0.25	0.5	1	1.5	2	3
y		3.34	2.82		1.67		1.15

(b) On the grid opposite, draw the graph of $y = 1 + 3e^{-x}$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$x = e^{-x} \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\ln(x-1)^3 = -3x \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(4)

OLevels.COM — SCHOOL —

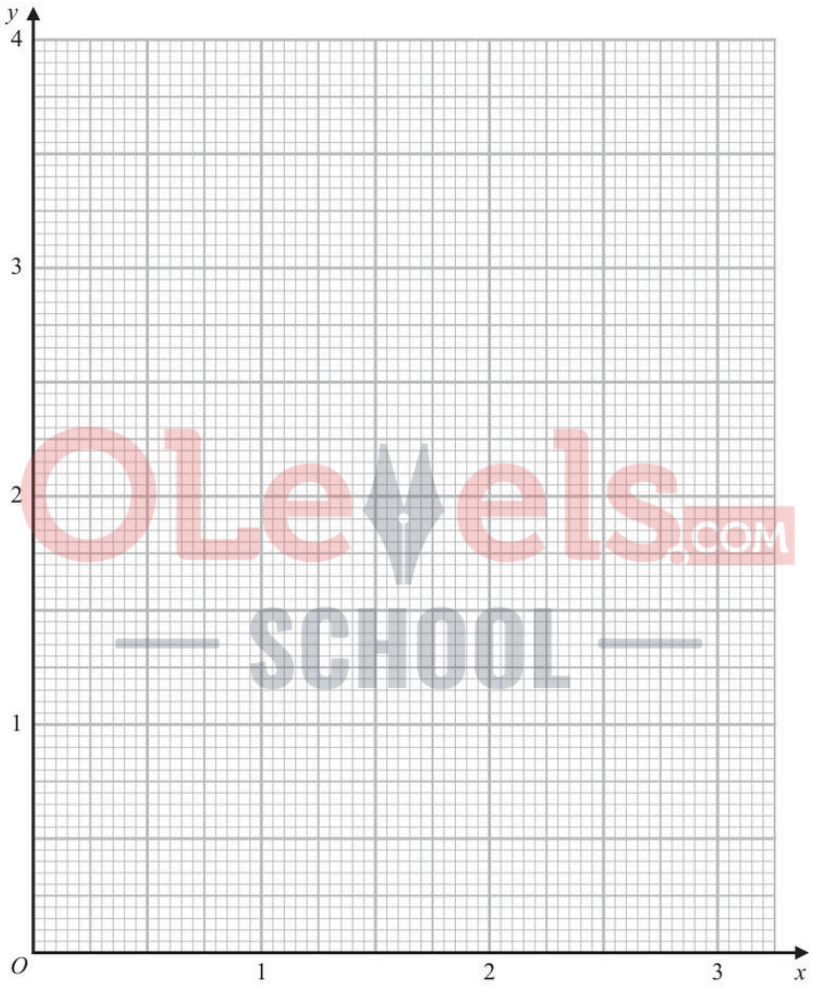
DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

10. Question 6 · part d · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

Question 6 continued



Turn over for a spare grid if you need to redraw your graph.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 1 6 6 1 A 0 1 7 3 2

17

Turn over

10. Question 6 · part d · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

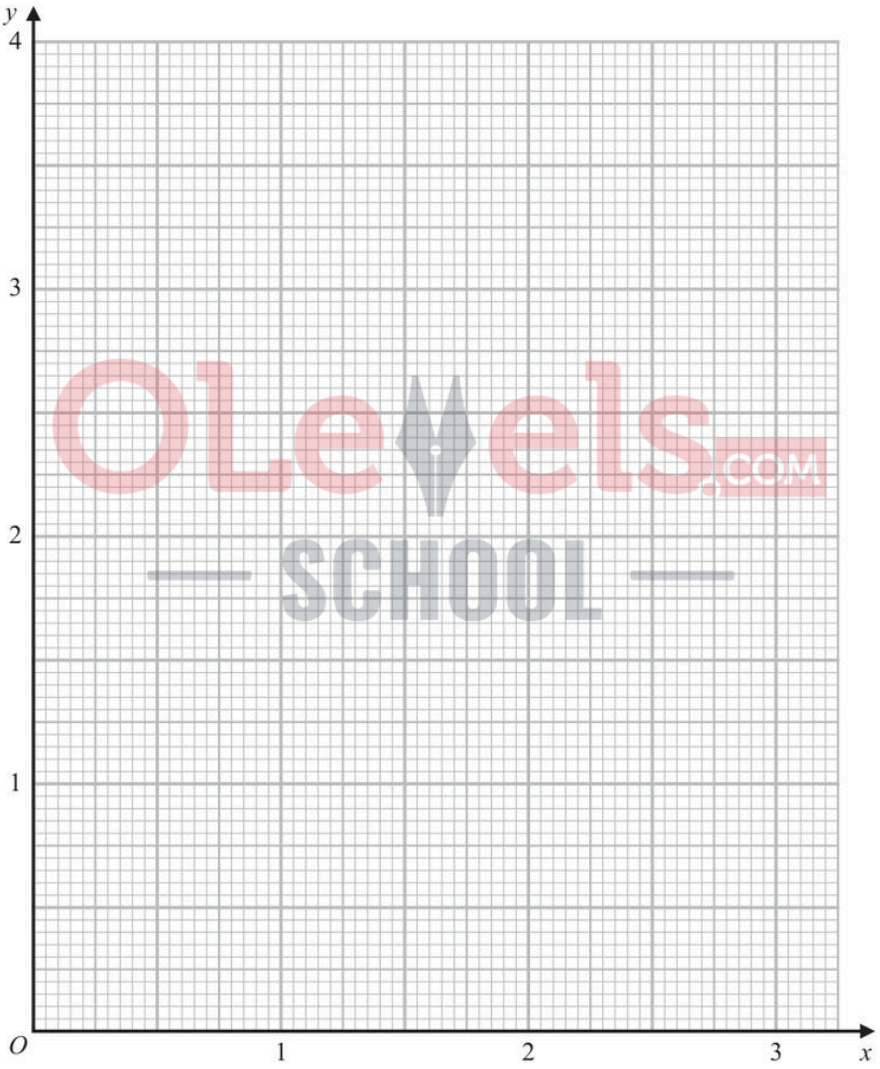
10. Question 6 · part d · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

Question 6 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(Total for Question 6 is 11 marks)

10. Question 6 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 4 · part c · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

4 (a) Complete the table of values for $y = \log_{10}(6x-1) - x$ giving your answers to 2 decimal places.

x	0.25	0.5	1	1.5	2	2.5	3
y		–0.20	–0.30	–0.60			

(2)

(b) On the grid opposite, draw the graph of $y = \log_{10}(6x-1) - x$ for $0.25 \leqslant x \leqslant 3$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$10^{\frac{3x-4}{2}} = 6x - 1 \quad \text{in the interval } 0.25 \leqslant x \leqslant 3$$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

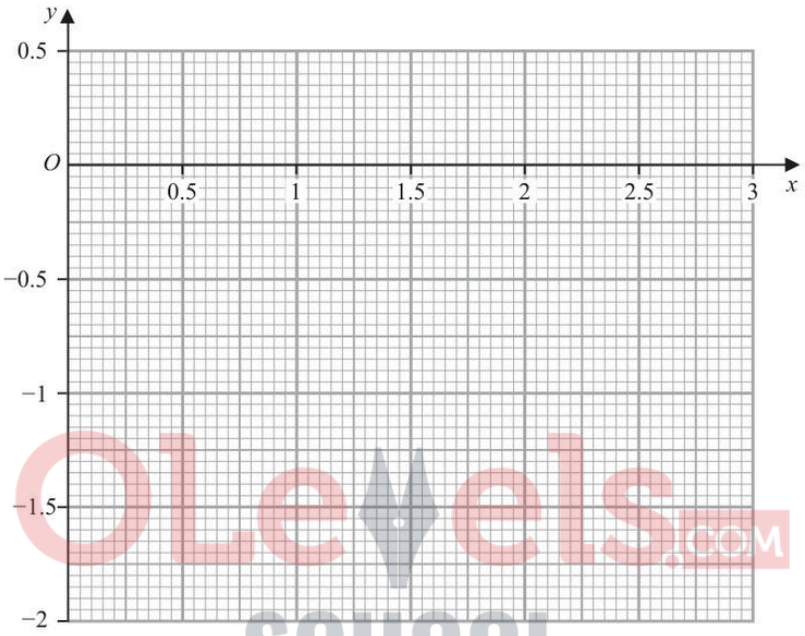
OLevels.com SCHOOL

11. Question 4 · part c · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Turn over for a spare grid if you need to redraw your graph.

11. Question 4 · part c · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

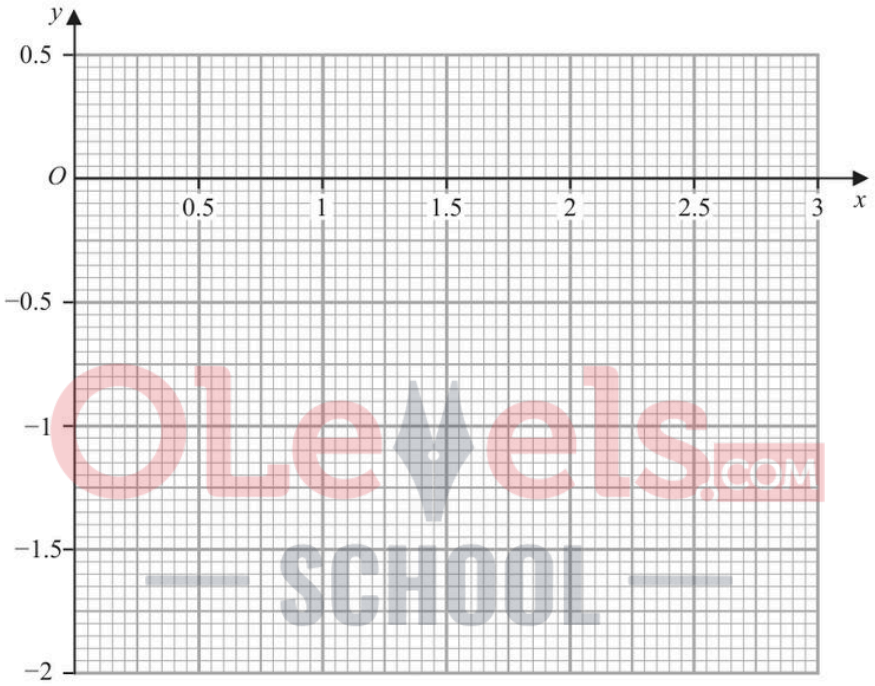
11. Question 4 · part c · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

(Total for Question 4 is 8 marks)

11. Question 4 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 7 · part c · 6 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 (a) Complete the table of values for $y = 3\log_3\left(x^2 - x\right) - 2x$ giving your answers to 2 decimal places.

x	1.3	1.5	2	2.5	3	3.5
y		−3.79	−2.11			

(2)

(b) On the grid opposite, draw the graph of $y = 3\log_3\left(x^2 - x\right) - 2x$ for $1.3 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$3^{\frac{4}{3}-x} = 9\left(x^2 - x\right)^3 \quad \text{in the interval } 1.3 \leqslant x \leqslant 3.5$$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

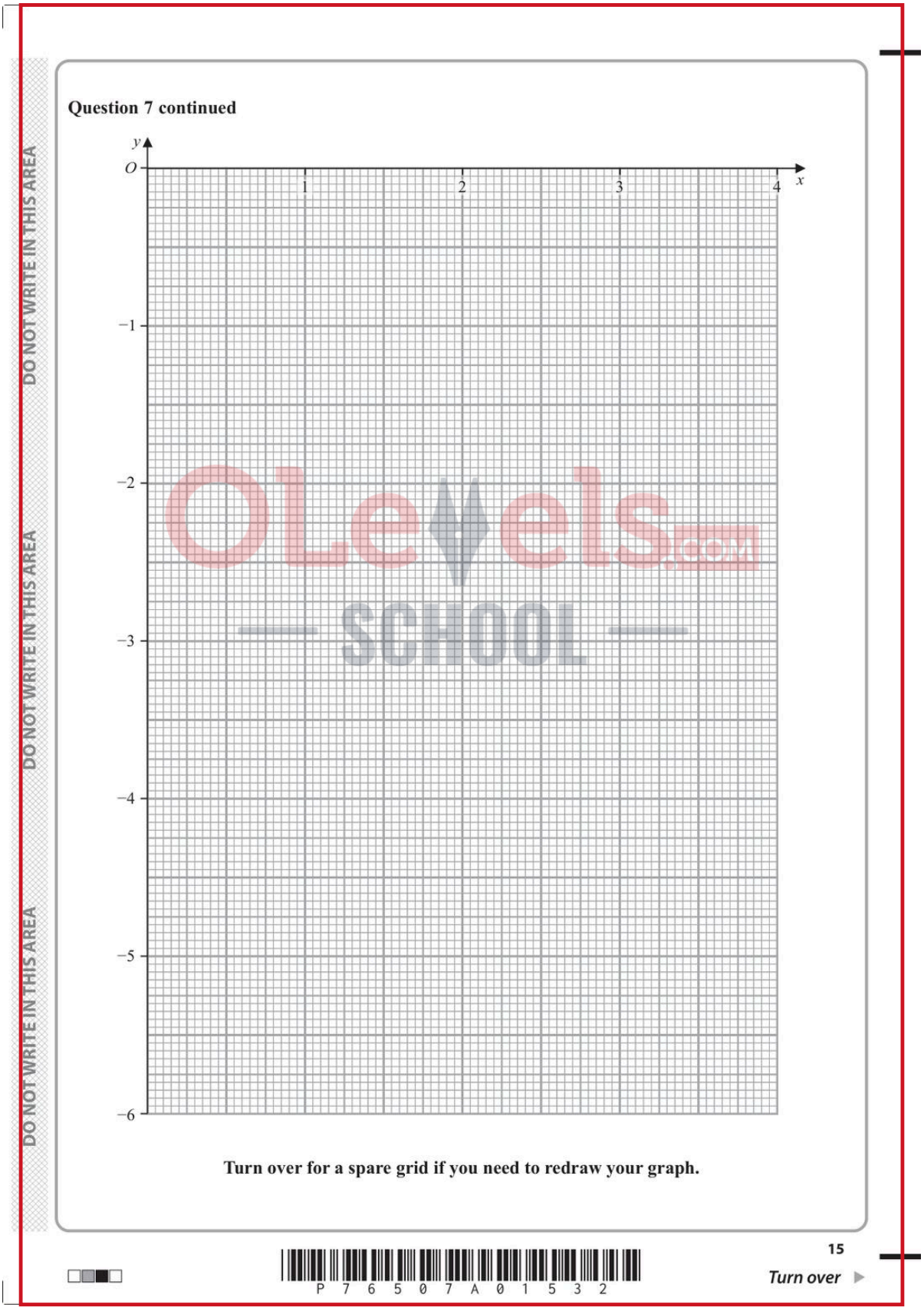
OLevels.com SCHOOL

P 7 6 5 0 7 A 0 1 4 3 2

12. Question 7 · part c · 6 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.



12. Question 7 · part c · 6 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

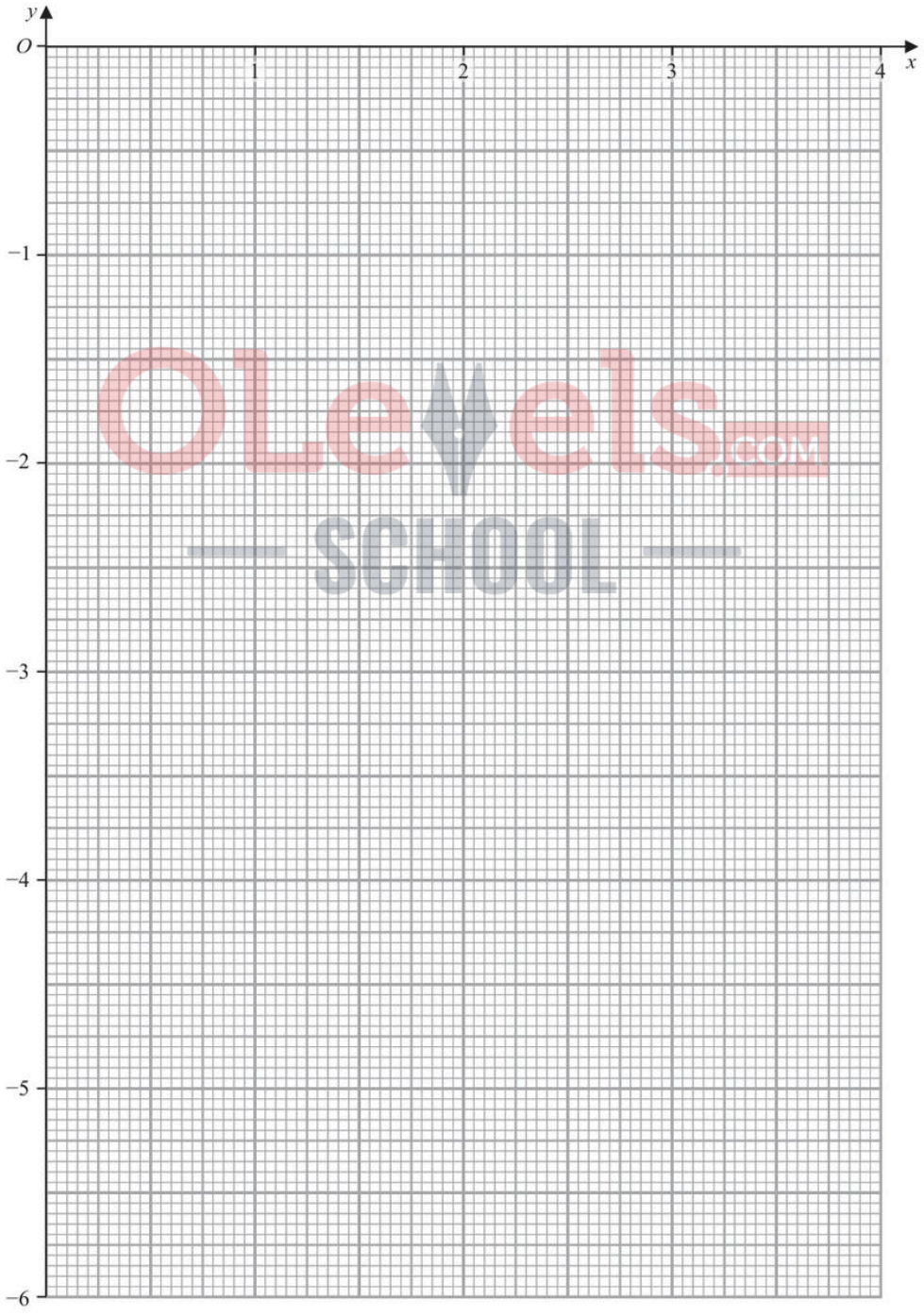
12. Question 7 · part c · 6 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

Only use this grid if you need to redraw your graph.



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

(Total for Question 7 is 10 marks)

12. Question 7 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL